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Integrated Vertical Power Delivery – Review & Challenges

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***Abstract*— This paper reviews and discusses technology advancements required for supporting integrated vertical power delivery in the package. By integrating switching devices, passives (inductors and capacitors) and embedded cooling solutions, this paper argues that high power delivery efficiencies are possible provided the appropriate power delivery architectures are used. The need for early-stage design space exploration along with a co-design framework are also discussed as being essential elements in the quest for achieving integrated vertical power delivery solutions.**

***Index Terms*—Integrated Voltage Regulator (IVR), vertical power delivery.**

I. Introduction

As the electronics industry transitions into the artificial intelligence (AI) era, significant power delivery challenges are emerging. In the past, electricity consumption by Data Centers in the United States was around 4.4% of the total energy generation. With the advent of AI data servers, this number is expected to increase to 12%. On a global scale, the rate of consumption is increasing 30% per year due primarily to AI [1]. What is alarming is that AI data centers are consuming energy at roughly four times the rate at which additional electricity is being added to the grids. Clearly, new power generation methods such as nuclear energy and others are necessary to support such a trend. In parallel, efficient system, chip, and software architectures need to be investigated to ensure that the least amount of power is being wasted.

The pre-2000 era saw the proliferation of single chip packages where the projected power levels of Central Processing Units (CPUs) was estimated to be 200 W in 2010 [2]. This changed quickly with the scaling of the transistor, where the CPUs and Graphical Processing Unit (GPUs) started consuming upwards of 600W of power. An example is Nvidia's H100 GPU introduced in 2022 at 700 W. Use of these processors in data centers has led to kilowatts of power consumption per rack in the past. With the continued scaling of the transistor and proliferation of AI, the new data centers are expected to consume Megawatts of power per rack, which represents around 3 orders of magnitude increase in power consumption.

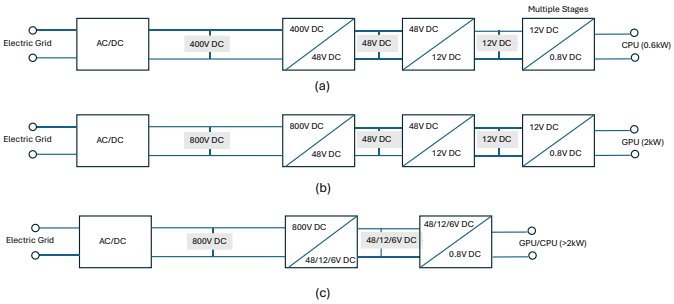


Fig. 1. Power Delivery Schemes: (a) Data centers with 0.6 kW CPU; (b) AI data centers with 2 kW GPU; and (c) AI data centers using heterogeneous integration with power levels exceeding 2 kW.

With the trend towards tighter integration, higher power densities need to be supported, requiring power distribution networks with low transmission losses. A power delivery system that is less complex is required, thereby making it more reliable. Three versions of power delivery schemes from the electric grid to the processor are shown in Fig. 1. In the legacy data centers, 400 V DC voltage goes through multiple stages of DC/DC conversion prior to providing the necessary power (voltage and current) to the CPUs. Multiple stages of conversion causes power delivery inefficiencies, takes up significant space in the racks for the power delivery units (PDU), making this approach not scalable beyond around 200 kW per computer rack. Hence, alternate power distribution schemes are being proposed primarily for AI data centers where 400 V is being replaced with 800 V, leading to lower transmission losses in the copper wires. The DC/DC conversion then follows the sequence 800 V-48 V-12 V-0.8 V, using fewer stages of conversion, thereby improving power delivery

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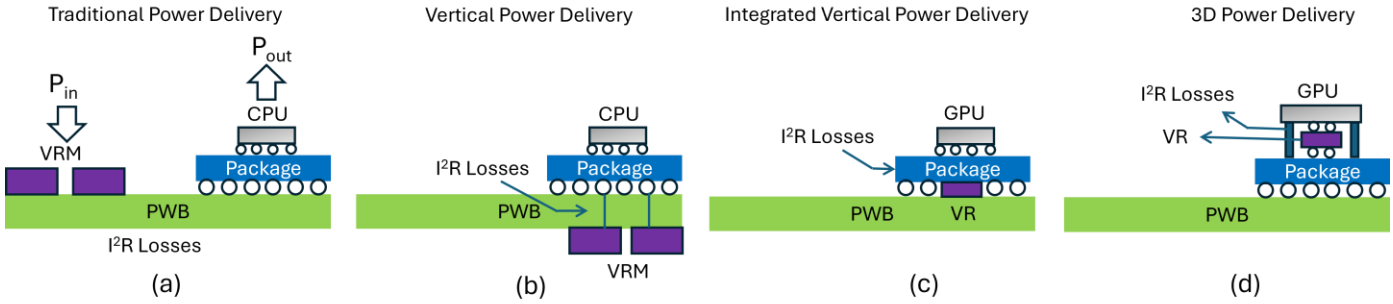


Fig. 2: (a) Traditional power delivery with multiple VRM stages; (b) Vertical Power Delivery with VRM attached beneath CPU on PWB; (c) VR integrated beneath the package referred to as integrated vertical power delivery; and (d) 3D power delivery with VR stacked beneath the chip.

efficiencies. The 800V DC voltage from the electric grid is expected to increase end-to-end efficiencies by around 5%, which is large given the power levels involved. This approach is expected to also lower the cost of thermal solutions [3]. As we move towards heterogeneous integration supporting upwards of 2kW on a single package substrate with multiple GPUs and CPUs, the conversion stages need to be reduced even further to improve efficiencies. Possibilities include 800 V – 48 V – 0.8 V; 800 V – 12 V – 0.8 V; or 800 V – 6 V – 0.8 V.

The pre-2000 era saw the use of discrete voltage regulator modules (VRM) to supply power often placed far away from the point of load CPU on the printed wiring board (PWB). That scenario is beginning to change with the introduction of Integrated Voltage Regulators (IVR) where fewer stages are being used for power conversion with integration occurring in the package [4].

II. Power Delivery Efficiency

In Fig. 1, as the power moves from the electric grid to the CPU and GPU, the role of packaging becomes critical in the last few stages, especially the conversion from 48 V to 0.8 V. In this context, the power delivery efficiency η is defined here as:

$$\eta = \frac{P_{out}}{P_{in}}, \quad (1)$$

with P_{in} representing the 48 V power supplied to the VRM and P_{out} being the power delivered to the chip at 0.8 V, as shown in Fig. 2. The VRM may comprise multiple cascaded conversion stages that supply and regulate the load as chip current demand varies. The objective is to ensure that minimum losses are incurred in the PWB. In traditional power delivery as shown in Fig. 2(a), the conversion from 48 V to 0.8 V occurs at the PWB and therefore large current flowing through the PWB causes I^2R losses, where R represents the resistance of the copper interconnects from the VRM to the chip, with a bulk of it in the PWB. To reduce routing losses, the VRM needs to be moved closer to the CPU. Three options are shown in Fig. 2: VRM attached below the chip on the PWB (Fig. 2(b)), Voltage Regulator (VR) integrated beneath the package (Fig. 2(c)), and 3D stacking where the VR is assembled beneath the chip using micro-bumps or other emerging assembly technologies such as hybrid bonding (Fig. 2(d)). VR represents the use of unpackaged chips, a major distinction from VRM that uses packaged chips and therefore represents a more discrete solution. Compared to traditional power delivery, where current flows horizontally, all three represent vertical power delivery schemes where the current flows vertically, and due to their short interconnect lengths, the resistance in the current path is small, thereby leading to higher power delivery efficiency. To differentiate these three approaches, we refer to them as vertical power delivery (VPD), integrated VPD (IVPD), and 3D power delivery. All three options require power conversion closer to the chip, along with its transmission to the chip making the implementation truly challenging.

VRs are essentially buck-derived switching regulator circuits requiring power field effect transistors, and storage components such as inductors and capacitors that do not typically scale easily making miniaturization difficult. The objective is to achieve efficiencies close to 100%. A detailed calculation of routing losses, single and multi-stage converters, along with some integration technologies are described in [4]. Option (d) has been analyzed in [5] where though the power losses can be reduced significantly, thermal coupling from the GPU to the VR can strongly impact the temperature of the VR die. We therefore extend the details in [4] with a focus on IVPD in this paper, which represents an emerging technology, where some of the thermal issues with 3D stacking can be resolved.

III. Heterogeneous Integration

The post 2022 era is expected to see a proliferation of heterogeneous integration (HI) solutions especially for AI data centers. The current emerging HI platforms integrate GPU with High Bandwidth Memory (HBM) on an interposer using high density interconnects. As new technologies emerge, we expect integration and miniaturization to continue leading to a future AI Data Center consisting of racks with optical switches and chiller systems for cooling as shown in Fig. 3. Each tray in the rack referred to as an AI server node, consists of GPUs, CPUs, and HBMs, tightly integrated with each other to ensure high bandwidth density and low energy per bit connectivity between chips. Such an integrated solution with embedded power delivery and cooling solutions will have needs that go far beyond today's technologies.

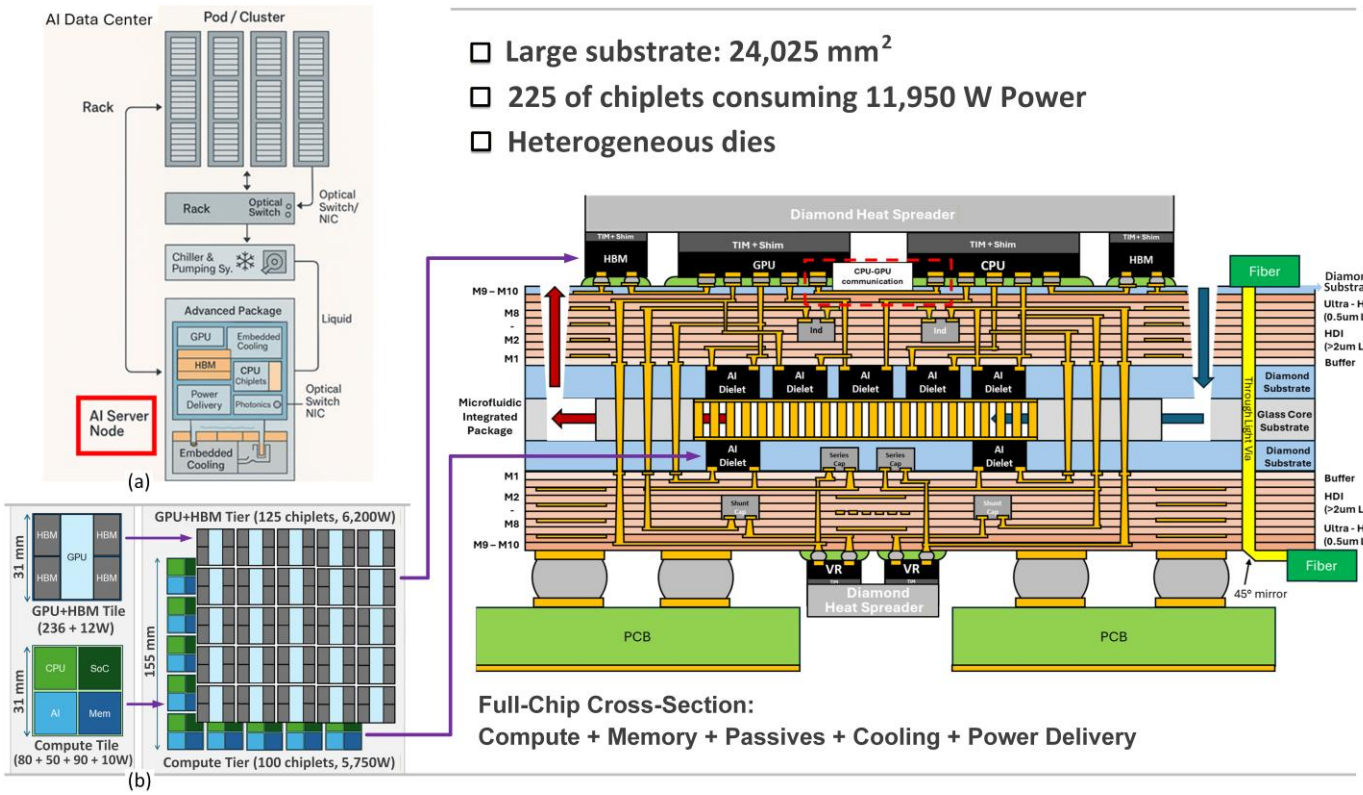


Fig. 3: A notional embodiment of a future AI data server node (Courtesy: Puneet Gupta (UCLA) and Sung kyu Lim (USC).

An embodiment of a notional and futuristic packaging solution is shown in Fig. 3 consisting of GPUs surrounded by HBMs on a tile, with multiple tiles placed side by side assembled on a large package substrate (>150 mm). Chips representing CPUs, new memory devices, system-on-chip (SoCs), and accelerators are embedded in the package with the VR assembled on the bottom side of the package and passive components embedded in the package. Both the chips assembled on top and bottom use flip chip technology thereby enabling heat removal from the top and bottom side. Integrated in the substrate are heat spreading materials, thermal interface materials, and microfluidics. In this embodiment the interposer and substrate are the same. The total power delivery that needs to be supported in the package is expected to be upwards of 11 kW. Such an embodiment requires IVPD to maximize power delivery efficiency. We provide a review and explore challenges for such emerging HI platforms as in Fig. 3 which includes power delivery architectures, design space exploration, minimizing parasitics, multi-physics modeling, inductor integration, embedded capacitors, and integrated cooling solutions. An example implementation on a glass substrate is illustrated to discuss the efficiencies achievable.

IV. Power Delivery Architecture

A. *High Step-Down IVR Architectures and Tradeoffs*

End-to-end power-delivery efficiency in data-center processors is determined by the combined impact of (i) conversion losses across one or more voltage-regulation stages and (ii) power distribution network (PDN) losses, which include copper routing losses, connector and plane/via resistance, and package/socket parasitics. As processors approach the kilowatt (kW) power range while rail voltages fall below 1 V, load currents reach the kiloampere (kA) regime and PDN losses, which is dominated by I^2R conduction, grow rapidly. This creates a central architectural tension: even if a high step-down VRM achieves high converter efficiency, a discrete implementation can suffer large board-level PDN losses between the VRM output and the processor package. Therefore, improving end-to-end efficiency requires not only reducing conversion losses but also integrating the regulation stage closer to the load (package-integrated or on-die).

From a conversion standpoint, a clear motivation emerges for single-stage or merged-stage [6]-[36] high step-down architectures operating from 12 V and 48 V where reducing the number of cascaded stages avoids the multiplicative penalty of per-stage losses and reduces inter-stage PDN overhead [4], [31], [36]-[38]. Table I tabulates this trend by summarizing a wide range of state-of-the-art high-conversion-ratio discrete/integrated architectures, including both single-stage and multi-stage topologies such as hybrid switched-capacitor converters (e.g., series- and flying-capacitor bucks), multi-level buck-derived configurations, and resonant-based architectures with and without transformers, which address the low-duty-cycle limitation of conventional buck switching converters at high step-down ratios. However, in kW/kA processors, converter-level improvements alone are insufficient because discrete implementations still deliver current through long board-level paths and connectors, incurring substantial I^2R losses after the VRM.

Table I.
Circuit Topologies and Integration Constraints of Power-Delivery Architectures for High Voltage IVRs (V_{in} > 5 V)

Note	Converter Topology (May Include Multiple Stages)	Single-Stage for V_{in} > 5V / No. Stages	Substrate / Integration Technology	Maximum Switching Frequency (f_{sw})[1]	**Vin** / Vout (V)	Full-Load Efficiency [2] (Output Power)	Power Delivery Scheme[3]	Inductance/ Magnetic Technology	No. Power Switch (Min.)	Integration Constraints as high voltage IVR (V_{in} > 5V)
2020, UofT [6]	Cross-Coupled Flying Capacitor Buck	Yes / 1	PCB/ Discrete	125 kHz	**48**/1	91.1% (40 W)	Traditional	1 µH/ *Discrete Ferrite*	8/ Si	Limited f_{sw}[8] (**hard switching**)
2024, UC Berkeley [7]	*1.* Hybrid Flying- Cap. *2.* Merged Series Cap.	No / 2 Merged	PCB/ Discrete	240 kHz	Total: **48**/1	Total: 86% (1500 W)	Traditional	260 nH/ *Discrete Ferrite*	56/ Si	Limited f_{sw}[8]
2020, UNIU-Google [8]	*1.* Switched-cap. divider. *2.* Three-level Buck *3.* Interleaved Buck	No / 3	PCB/ Discrete	*1st*: 205 kHz *2nd*: 520 kHz 3rd: 520 kHz	**54**/7.2 **7.2**/1.8 **1.8**/1	*1st*: 84% (360 W) *2nd*: 94.3% (180 W) 3rd: 94% (170 W)	Traditional	1 µH/ *Discrete Ferrite*	20/ Si	Limited f_{sw}[8]/ discrete front-end stages
2020, Bel Power [9]	Isolated Full-Bridge buck	Yes / 1	PCB/ Discrete	242 kHz	**48**/1	90.5% (70 W)	Traditional	*N. R. (Transformer)*	6/ Si	Limited f_{sw}[8]/ Transformer
2020, UC Berkeley- [10]	Hybrid Switched-Capacitor Buck	Yes / 1	PCB/ Discrete	250 kHz	**48**/1	86.4% (65 W)	Traditional	600 nH/ *Discrete Ferrite*	10/ Si	Limited f_{sw}[8]
2022, UC Berkeley [11]	Hybrid Series Capacitor Buck (*Dickson*[2]*-PoL*)	No / 2 Merged	PCB/ Discrete	280 kHz	Total: **48**/1	Total: 88.4% (270 W)	Traditional	300 nH/ *Discrete Ferrite*	22/ Si	Limited f_{sw}[8]
2019, CU Boulder [12]	Hybrid Flying-Capacitor Buck	Yes / 1	PCB/ Discrete	300 kHz	**48**/1	80.4% (40 W)	Traditional	2.2 µH/ *Discrete Ferrite*	9/ Si	Limited f_{sw}[8]
2021, Analog Device [13]	Dual output (Non-isolated)	Yes / *N. R.*	µModule/ Integrated	350 kHz	**48**/1	88.1% (50 W)	Traditional	*N. R.*/ Package Integr.	N. R.	Limited f_{sw}[8]
2021, Princeton-pSemi [14]	*1.* Flying-Capacitor 2. Series Capacitor	No / 2	PCB/ Discrete	417 kHz	Total: **48**/1	Total: 89.1% (450 W)	Traditional	50 nH/*Coupled Discrete Ferrite*	42/ Si	Limited f_{sw}[8]
2016, BGU [20]	Double Series Capacitor	Yes / 1	PCB/ Discrete	500 kHz	**48**/1	83.9% (30 W)	Traditional	440 nH/ N. R.	5/Si	Limited f_{sw}[8]
2020, CU Boulder [15]	*1.* Hybrid Flying-Cap. *2.* Interleaved Buck	No / 2	PCB/ Discrete	*1st*: 125 kHz / *2nd*: 500 kHz	Total: **48**/1	*1st*: 97.5% (90 W) *2nd*: 87.1% (90 W)	Traditional	120 nH *Planar Ferrite-*	18/ Si	Limited f_{sw}[8]
2020, CPES VT [16]	*1.* ZVS LLC Resonant Trans. (Soft Switched) *2.* Buck Regulator	No / 2 Parallel	PCB/ Discrete	600 kHz	**48**/1	Total: 92.5% (80 W)	Traditional	36 µH: L_m[3] 190 nH: Ind./ *Planar Ferrite*	20/ Si	Transformer losses and integration
2022, UC Berkeley [17]	Hybrid Flying-Cap. Buck (*Symmetric Dicks.*)	Yes / 1	PCB/ Discrete	750 kHz	**48**/1	71.5% (105 W)	Traditional	150 nH/ *Discrete Ferrite*	14/ Si	Limited f_{sw}[8]
2024, CPES VT [18]	*1.* ZVS LLC Resonant Trans. (Soft Switched) 2. Buck Regulator	No / 2 Parallel	PCB/ Discrete	*1st*: 840 kHz *2nd*: ----	**48**/1.8 **1.8**/0.8	*1st*: 91.8% (450 W) *2nd*: *N. R.*	Traditional	36 µH: L_m[4] 190 nH: Ind./ *Planar Ferrite*	28/ Si-GaN	Transformer losses and integration
2022, Princeton-Intel-Google [19]	*1.* Hybrid Flying-Cap. *2.* Interleaved Buck *(LEGO-POL)*	No / 2 Merged	PCB (Stacked)/ Discrete	*1st*: 286 kHz *2nd*: 1 MHz	**48**/8 **8**/1	Total: 85.7% (450 W)	VPD	85 nH/ *Discrete Ferrite*	40/ Si	Limited f_{sw}[8]
2024, UC Berkeley [21]	Hybrid Flying Cap. Buck (*Symmetric Dicks.)*	Yes / 1	PCB/ Discrete	1 MHz	**48**/1	88% (50 W)	Traditional	78 nH/ *Discrete Ferrite*	16/ GaN	Limited f_{sw}[8]
2020, Vicor [22]-[23]	*1.* 4switch ZVS Buck-Boost [17] *2.* Isolated Curr. Multiplier	No / 2	IC/ Integrated	1025 kHz	Total: **48**/1	Total: < 90.1%: 214 W [5]	Traditional	*N. R.*/ *Package Integrated*	10 Min./ Si	Transformer losses and integration
2023, Princeton-Intel-Google [24]	*1.* Hybrid Flying Cap. *2.* Interleaved Buck *(Mini-LEGO)*	No / 2 Merged	PCB (Stacked)/ Discrete	*1st*: 433 kHz *2nd*: 1.5 MHz	**48**/8 **8**/1	Total: 84% (240 W)	VPD	85 nH/ *Discrete Ferrite*	40/ Si	Limited f_{sw}[8]
2023, ASCEN (SRC)-Intel [25]	Hybrid Flying-Cap. *(3-level Hybrid Dickson*	Yes /1	Silicon/ 0.18 µm - Disc. Passive	1 MHz	**12-48**/1	73% (12 W)	Traditional	620 nH/ *Discrete Ind.*	11/ Si-GaN	Large passives/ Limited f_{sw}[8]
2016, Texas Ins. [26], [27]	Two-Phase Series Capacitor Buck	No / 1	PCB/ Discrete	2-5 MHz	**12**/1.2	79% (5MHz, 12 W) 84% (2MHz, 12 W)	Traditional	100 nH: 5MHz *Chip Inductor*	4/ Si	Limited f_{sw}[8]
2023, ASCEN-SRC-Intel [28]	Hybrid Flying-Cap. *(Series-parallel hybrid)*	Yes / 1	Silicon/ 0.18 µm - Disc. Passive	2-4 MHz	**12-48**/1	70% (12-Vin) 77% (48-Vin) (2MHz, 9.5 W)	Traditional	400 nH/ *Discrete Ind.*	7/ GaN	Limited f_{sw}[8]
2025, ZJU [29]	*1.* Series Capacitor Buck *2.* ZVS Resonant Switch-Capacitor	No / 2	Silicon/ 65 nm-CMOS (2nd stage)	*1st*: 1 MHz *2nd*: 9 MHz	**12**/1.8 **1.8**/1	Total: 84% (12 W)	Traditional	1st : $L_{1,2}$: 1µH/ *Discrete* 2nd : L_r = 1nH	8/ Si	Additional discrete front-end stage
2024, Intel [30]	Buck (Multi-phase)	No / 1	Silicon/ 55nm-BCD	3-10 MHz	**5.4**/1	87% (10MHz, 20 W) 89.4% (5MHz, 80 W	IVPD	17-55 nH/ *Package embe. Air Core*	8/ Si	Multiple discrete front-end stages
2025, Penn State CHIMES (SRC)[31],[32]	ZVS BCM Series-Capacitor Buck (Soft Switched)	Yes / 1	Glass/ Integrated [30] PCB/Disc. [31]	0.5-20 MHz Nom.: 5 MHz	**12-48**/1	88.5% (5MHz, 100 W, 12V-Vin[6]	IVPD [30] Trad. [31]	1.85nH (5MHz)/ *Discre-Emb.)*	6/ GaN	Inductor AC copper losses
2025, UCLA-CHIMES (SRC) [33]	Conventional Buck	No[7] / 1	Silicon / 0.13 µm (Cu-Cu 8-µm)	40 MHz	**12**/1	< 70% [6] *(Power: N. R.)*	IVPD	*N. R.*/ *air-core spiral*	2/ Si	Limited f_{sw}[8]/ discrete front-end stages
2019, Intel [34]	Split-Input Stacked Buck (Light load- ZVS)	No / 1	Silicon/ 14 nm (FIVR)	70 MHz	**1.6**/1	81% (0.5 W)	IVPD	2.5 nH/*Package embe. Air Core*	2/ Si	Multiple discrete front-end stages
2022, Intel [35]	Buck (Multi-phase)	No / 1	Silicon/ FIVR (3D-stacked)	90-140 MHz	**1.8**/1	85.5% (30 W) 600 W: parallel buck	3D Power Delivery	1.2-2.5 nH/ Coaxial Magnet *embedded* [34]	4/ Si	Multiple discrete front-end stages

[1]For multi-stage converters, switching frequencies are listed from the high-voltage stage to the low-voltage VRM stage.

[2]Overall full load efficiency for multi-stage converters is calculated as the product of individual stage efficiencies; gate losses are excluded, and air cooling is assumed.

[3]The Power Delivery Schemes represented in Fig. 2 include Traditional Power Delivery, VPD, IVPD, and 3D Power Delivery.

[4]L_m: transformer magnetizing inductance; Ind.: buck output-filter inductor.

[5]Only peak efficiency is reported (including gate-driver losses); full-load efficiency is therefore lower than the reported 90.1%.

[6]Theoretical and simulation results.

[7]Marked "No" because a conventional buck operated directly from 12-V relies on a very small duty cycle, resulting in poor conversion efficiency.

[8]Hard-switched converters require limited switching frequencies to minimize switching losses and maintain high efficiency.

high step-down topologies can be integrated (i.e., scaled to high switching frequencies (f_{sw}) to enable passives integration) while maintaining high efficiency at high input voltage (12–48 V)?

B. Loss Mechanisms and Integration Constraints at kW/kA Scale

Integration fundamentally pushes architectures toward multi-MHz switching to shrink passive component sizes, most critically the inductor; so that magnetics can be embedded in the package substrate and the converter can be placed at or within the processor package. Table I shows that most high-V_{in} hard-switched solutions are constrained to relatively low f_{sw} (typically hundreds of kHz to ~1–2 MHz) at 12–48 V because switching losses scale strongly with input voltage and switching transition energy. This f_{sw} limitation forces the use of larger inductances (hundreds of nanohenries to microhenries) requiring ferrite-core magnetics, which introduces additional core-losses as f_{sw} increases. In other words, many hard-switched high step-down architectures (e.g., [6]-[15], [17], [19]-[21], [24]-[28]) successfully improve the conversion ratio and can demonstrate attractive converter efficiency, but they are not compelling IVR candidates because they cannot sustain the high-f_{sw} operation required for passive integration at high input voltages.

Notably, recent fully integrated multiphase designs operating at tens to hundreds of MHz (e.g., Intel FIVR-class implementations) achieve sub-10-nanohenry inductance using package-embedded air-core or coaxial magnetics [39] by pushing f_{sw} very high; however, their input voltage is inherently limited to approximately 1.6–1.8 V (with a maximum f_{sw} of 140 MHz) [34], [35] or up to about 5 V (with a maximum f_{sw} of 10 MHz) [31]. A fully integrated 12-to-1 V, 40-MHz buck architecture reported in [33], extends this approach using a conventional buck topology enabled by advanced integration technology (e.g., Cu–Cu bonding) but incurs very high conversion losses due to the extremely low duty cycle and elevated switching and conduction losses associated with 12-to-1 V buck conversion. Scaling these architectures to 12–48 V while maintaining high efficiency requires additional discrete front-end stages and/or a reduction in f_{sw} with a corresponding increase in inductance, thereby diminishing the benefits of IVR integration.

C. ZVS Architectures for High Efficiency kW IVRs

This is where Zero-Voltage-Switching (ZVS)/soft-switching becomes a key differentiator [31]. By reducing/eliminating switching loss, ZVS architectures relax the high-V_{in} frequency ceiling and enable higher f_{sw} while preserving efficiency. As summarized in Table I, several soft-switched families such as LLC-based resonant converters [16], [18], resonant switched-capacitor concepts [29], and ZVS four-switch buck–boost architectures [22] demonstrate this principle. However, many resonant high step-down approaches require additional regulation stages (series or parallel) to achieve tight output regulation, while transformer-based implementations introduce increased magnetic volume and integration complexity, limiting their suitability for compact package-integrated IVR architectures. In addition, the four-switch buck–boost topology reported by Vicor cannot independently provide a high conversion ratio and is commonly paired with a transformer-based current-doubler stage [23], further complicating system-level integration.

In contrast, the ZVS boundary-conduction-mode (BCM) series-capacitor buck in [31], [32] directly targets the integration bottleneck: it maintains high-V_{in} capability (12–48 V) while enabling multi-MHz operation with ultra-low inductance (~2 nH at 5 MHz), making package-level integration practical without requiring an added regulation stage. Although ZVS-BCM operation can increase conduction-related losses, the ability to operate efficiently at high f_{sw}, enabled by soft switching operation combined with substantially reduced board-level PDN losses through IVPD-based integration, creates a stronger end-to-end efficiency pathway than hard-switched architectures in kW/kA regimes.

Overall, the comparison in Table I supports a step-by-step conclusion: (1) reducing conversion stages (single/merged high step-down architectures) is essential to avoid compounding conversion losses, (2) kW/kA operation makes board-level PDN losses a dominant penalty unless VR is integrated near the load, (3) integration demands multi-MHz switching to shrink magnetics, especially inductors, yet (4) high-V_{in} hard-switched designs are frequency-limited by switching loss and therefore struggle to become true IVRs, and (5) ZVS-enabled high-V_{in} architectures provide the most viable route to simultaneously achieving high f_{sw}, small valued passives that can be integrated, leading to high efficiency, enabling package-level integration and meaningful end-to-end efficiency gains.

V. Design Space Exploration

Early-stage exploration of VPD architectures is essential for identifying configurations capable of achieving high end-to-end efficiency under kilowatt-scale power and multi-A/mm² current-density constraints. At this stage, the goal is system-technology co-optimization (STCO), which focuses on identifying dominant loss mechanisms (e.g., horizontal and vertical PDN loss and VR losses) and understanding their tradeoffs across different VPD configurations.

A representative distributed VPD system is shown in Fig. 4 with multiple VRs embedded beneath the load die and the corresponding end-to-end loss breakdown for several architectural options. The top and side views illustrate spatial VR distribution and vertical interconnects (e.g., Through Silicon Vias TSVs/Through Glass Vias TGVs/Microvias), while the stacked bar chart decomposes total loss into supply-to-VR routing (horizontal PDN), VR-to-POL routing (vertical PDN), and VR losses (including internal VR routing, inductor conduction, power-switch conduction and switching losses).This breakdown clarifies how architectural choices shift loss dominance among horizontal and vertical PDN and VR losses.

In conventional architectures with 48 V-to-near-point of load (PoL) conversion at the PWB level, horizontal PDN losses dominate because high current must be routed laterally through the package and interposer. Architecture-level analyses [40], [41] show that for representative ≥ 1 kW loads, horizontal PDN losses alone can reach several hundred watts under low-voltage, high-current delivery, rendering such configurations impractical for advanced heterogeneous systems. This motivates higher-voltage vertical delivery with major conversion deferred closer to the load.

As shown in Fig. 4 and consistent with [40], [41], single-stage, high-ratio on-package conversion suppresses horizontal PDN loss but shifts dissipation into compact VRs operating at elevated switching frequency. Vertical PDN loss remains secondary yet non-negligible due to the finite resistance of micro-bumps and TSVs/TGVs. This challenge is further exacerbated in ultra-high-voltage (> 48 V) schemes. The loss comparison in the figure is based on a buck-derived switched-capacitor VR architecture operating in multiphase CCM [20]. The switching frequency for the individual power conversion schemes is selected to minimize total VR loss under ripple and density constraints, resulting in an optimized operating range of approximately 3-4 MHz (see Fig. 4). The reported losses include all switching loss components (including gate-drive) as well as power switch, inductor winding, and internal VR routing conduction losses. Although inductor core and capacitor ESR losses are excluded, under multiphase CCM operation, ESR dissipation is expected to be negligible. Mitigation of core losses requires advanced low-loss magnetic materials and device-level innovations, as discussed in Section VI.B.1.

Multi-stage and distributed IVPD (DIVPD) architectures further reshape this balance by partitioning voltage conversion and load current across multiple regulators. Increasing the number of VRs reduces per-regulator current, lowering conduction losses and mitigating localized thermal concentration. DIVPD also enables dynamic power management, reduces thermal gradients and local stress, and alleviates warpage risks. With internal VR losses dominating total dissipation, a primary design concern becomes the tradeoff between power-switch switching losses and overall internal VR conduction losses. As conversion ratios and switching frequencies increase, careful co-optimization of VR components is required to balance switching and conduction losses and minimize overall VR dissipation. In highly distributed architectures, the large number of switching devices can render aggregate gate-drive loss a non-negligible contributor, even when individual devices are downsized through current partitioning. These further underscores the importance of system-level and VR co-optimization of switching frequency, device sizing, and phase count under thermal, footprint, and current-density constraints. As conversion ratios and f_{sw} increase, careful co-optimization of VR components is required to balance switching and conduction losses and minimize overall VR dissipation.

As illustrated in Fig. 4, DIVPD enables efficiencies above 80%, trending toward the high-80% range, provided power-

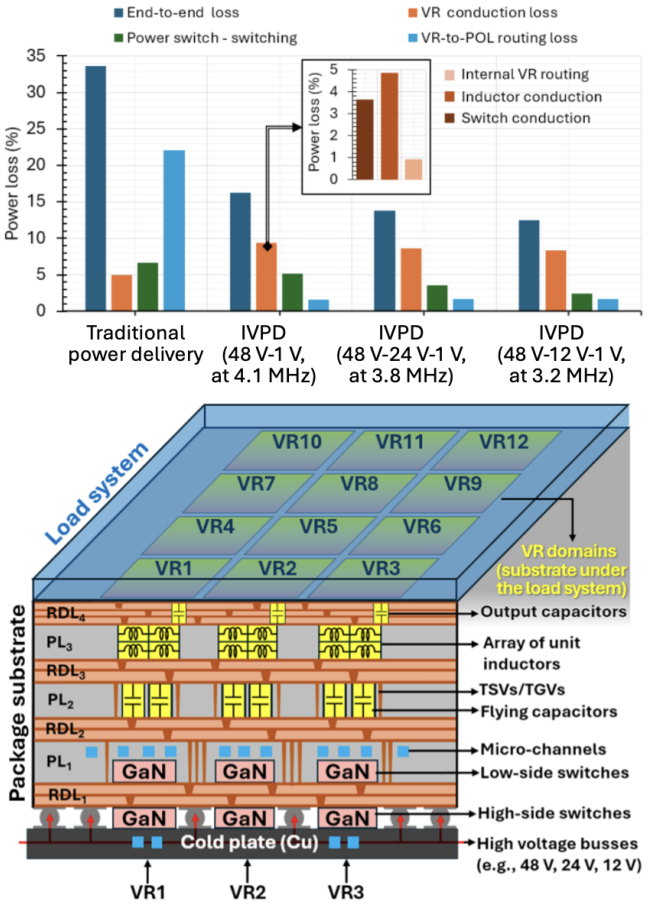


Fig. 4. Illustration of how VPD architecture impacts loss distribution. IVRs placed beneath the load die shorten horizontal current paths, while multi-stage conversion redistributes loss from PDN routing to VR, enabling higher end-to-end efficiency.

switch sizing, regulator placement, interconnect density, and switching frequency are co-optimized [40], [41]. These efficiency projections reflect what is theoretically attainable under coordinated co-optimization, rather than a mature, deployable design solution. Practical realization demands scalable, architecture-aware modeling and design-automation frameworks that can handle the complexity of tightly coupled electrical–thermal–mechanical constraints.

These trends suggest several design guidelines: (1) Horizontal PDN losses should be minimized by preserving a higher distribution voltage deep into the stack, limiting lateral high-current routing; (2) Ecessively high single-stage conversion ratios must be carefully managed to avoid shifting the bottleneck to VR switching and conduction losses; (3) Distributed and multi-stage regulation effectively balances VR, vertical PDN, and horizontal PDN losses, enabling scalable efficiency improvements with increasing power density. Overall, IVPD efficiency is governed not by a single loss component but by the coordinated tradeoff among these mechanisms, underscoring the importance of architecture-level co-optimization for next-generation high-power systems.

VI. Design and Implementation Challenges

Under high current and high f_{sw} operation, the physical placement of active devices, passives, and interconnects become an important design variable, directly impacting efficiency, current density, electromagnetic interference (EMI), and thermal performance. In this context, IVPD represents an increasingly necessary approach for implementing high-input-voltage regulators in advanced data-center processors with current densities approaching 5 A/mm² [4], [31], [38], although they introduce significant practical implementation challenges.

In a representative stacked configuration (Fig. 4), the high-side and low-side switches are placed at the bottom of the stack, closest to the input supply, while embedded high-voltage series/flying capacitors, inductors, are distributed vertically, with the PoL located at the top, directly interfacing with the processor power pins. This vertical current flow is compatible with both conventional high-current IVRs and distributed power delivery architectures [42], [43] and significantly reduces lateral current spreading and associated resistive and inductive parasitics.

A. Minimizing Parasitic Loops

Beyond resistive losses, the reduction of high-frequency parasitic inductive loops is a key motivation for integrated IVR design. Vertical integration enables a substantial reduction in power-loop inductance, which is essential for achieving high f_{sw} at high current by suppressing switching-node ringing. Fig. 5 illustrates a representative three-phase series-capacitor buck IVR architecture, highlighting three dominant high-frequency commutation loops that require ultra-low parasitic total inductance to minimize ringing at multi-megahertz f_{sw}. By vertically stacking high-voltage series capacitors (C_{s1} and C_{s2} in Fig. 5) directly above the flip-chip switching devices, both device package inductances and interconnect inductances can be significantly reduced, enabling higher f_{sw} while limiting voltage overshoot associated with large di/dt and dv/dt transients. Furthermore, advanced interconnect technologies, such as copper–copper bonding for chip embedding in 3D integration, can further reduce interconnect parasitics and enhance these benefits as employed in the IVPD architecture in [33].

B. Passives Integration

In buck-derived IVR architectures, the inductor is inherently located on the low-voltage side, near the output capacitor and the load, as shown in Fig. 5. Accordingly, in a vertically stacked system, the inductor is ideally placed toward the top of the stack, close to the PoL, to minimize conduction losses and output-loop parasitics, as illustrated in Fig. 4. Although this placement is advantageous from an electrical perspective, it introduces additional challenges related to magnetic material integration, thermal coupling to the processor, mechanical stress, and material compatibility. In addition, high transfer-ratio buck-derived architectures inherently require larger inductance values than low-voltage IVRs. When combined with high output current requirements, the realization of suitable

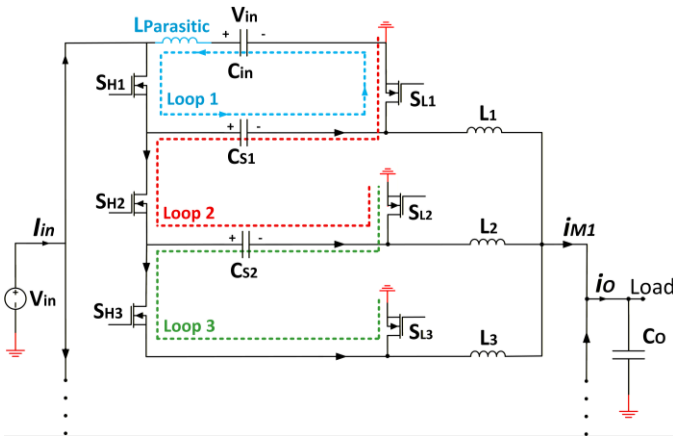


Fig. 5: Example of high-frequency power loops in a three-phase series-capacitor buck architecture requiring low equivalent parasitic inductance loop at multi-megahertz switching frequencies.

inductors within a stacked integrated architecture becomes significantly more challenging.

Similarly, in single-stage high-input-voltage IVRs, the integration of high-voltage series or flying capacitors presents another major challenge, as these components must simultaneously satisfy breakdown voltage, high capacitance density in a low-profile form factor, low equivalent series inductance and resistance (ESL/ESR), and high self-resonant frequency (SRF) requirements.

While vertical stacking enables significant improvements in switching performance and power density compared to traditional two-dimensional implementations, it also shifts the primary bottleneck toward the realization of embedded passive components under stringent voltage, current, and integration constraints. These challenges motivate a detailed examination of embedded inductors and capacitors.

1) Inductors

Inductors store magnetic energy and provide current to the PoL. From an integration standpoint, inductors represent the most difficult passive to embed in the package for several reasons. In discrete form, their height is upwards of 1.8 mm for the values required, which is much thicker than the package and therefore is not feasible to embed. Thin-film large-valued inductors require magnetic materials for miniaturization, leading to the electrical performance being affected by their conductor and magnetic core loss. High permeability magnetic materials used as the core in general have high magnetic loss tangent. Increasing saturation current often requires the use of air gaps further increasing inductor size. Small-valued inductors are easier to integrate where an air-core can be used which eliminates core loss and increases saturation current. However, small-valued inductors are possible only with very high f_{sw} (> 100 MHz) and therefore are limited to low conversion ratio IVR (1.7 V/1 V). For high conversion ratio IVRs as described in this paper, the objective should be to investigate architecture that reduce the inductor values as much as possible, thereby requiring magnetic materials with lower

Table II.
Performance Comparison of Embedded and In-Package Inductors for High-Frequency IVRs

Note	Core Material (Permeability: μ_r)	Substrate	Structure	Ind. (nH)	f_{sw} (MHz)	R_{dc} ($m\Omega$)	Peak | Sat. Current*	Thickness (mm)/ Current Density (A/mm²)	FOM (W) ($L.I_{Peak}^2.f_{sw}$)
2018, DTU [44]	Air (1)	FR4	2D Spirals Planar+	685	2	40.7	1A	1.1 / 0.0044	1.37
2021, TSMC [45], [47]	Air (1)	RDL	3D Rectangular (InFO)	2.1	200	3.2	0.8 A	0.15 / 0.26	0.27
2016, Intel [46]	Air (1)	Organic flip-chip	3D Solenoid	1-6.7	140	6.7-36.3	≥ 2.5 A**	0.2 - 0.8 / 1.04	5.86
2023, UESTC [61]	Air / Organic Filler	Glass	3D spiral (TGV)	11.1	4000	N. R.	RF App.	Area: 0.164 mm²	N. R.
2025, UIC [64]#	Air (1)	Dielectric	2D Spirals Planar	20	100	11	10A (Sim.)	0.5 / 1.42	No fabrication#
2021, Intel [39]	Composite (8.5)	RDL	3D Solenoid (Coax MI)	2.5	90	12	8 A	N. R. / 20	14.4
2020, GT (SRC) [49]	Composite (90)	FR4	3D toroidal	374	10	24	2.5 A	0.54 / 0.41	23.4
2021, GT (SRC) [50]	Composite (24)	FR4	3D toroidal	60.7	10	22.8	5 A	0.6 / 1	15.18
2020, Intel [51]	Composite	Package flip-chip	3D Solenoid	3	100	12	4	0.2 / 8	4.8
2022 GT (SRC) [52]	Composite (43)	FR4	3D toroidal (IND3)	68-600	2-10	49-89	2.5	0.4 / 0.28-2.5	4.25
2025, GT (SRC) [60]	Composite (180)	FR4	2D Spiral (Dual core)	330	2	85	5 A	0.55 / 0.56	8.1
2019, Shinshu-U[53]	Fe Composite (9)	FR4	3D spiral	150	20	N. R.	0.55	2.3 / N. R.	0.91
2020, Tyndall [54]	Fe Composite	FR4	3D toroidal	780	2	66	4	0.4 / 0.046	24.8
2017, DTU [55]	Fe Composite (2.4)	Silicon (TSV)	3D toroidal	85	12.5	230	1.6	0.28 / 0.286	2.72
2012, CU[56]	Thin film laminated	Silicon	Solenoid+	18.4	100	93	4	N. R. / 11	29.4
2020, GF-Wuerth [57]	Thin film (800)	Silicon	3D Solenoid	120	15	270	0.4	N. R. / 0.43	0.29
2015, Ferric-TSMC [58]	Thin film (500)	Silicon	Solenoid	212	150	1130	0.45	0.03 /11	6.44
2020, TSMC [59]	Thin film (N. R.)	Silicon	3D Solenoids	3.9	120	N. R.	4	N. R.	7.5
2024, CAS [62]	Thin film (Ferrite)	Silicon	3D solenoid+	339.5	10	308	2.2	0.2 / 0.37	16.4

+This structure is a two-winding coupled inductor.

*For air-core inductors, winding conductor loss and temperature rise limit the RMS/peak current. Magnetic-core inductors are generally limited by core saturation at currents below the copper thermal limit.

**Current per inductor: Parallel inductors are used to reach currents above 20 A.

#Only simulation results (Not fabricated).

relative permeability. This can enable increased saturation current due to the material property and reduced loss. Commonly used inductor topologies are described in [4]. Embedded inductors require that the magnetic field be confined inside the core to minimize electromagnetic interference (EMI), placing restrictions on the types of inductor topologies that can be used [4].

In Table I [13], [20], [25], [28], and [29], discrete inductors are used, which achieve lower DC copper resistance, reduced core losses, and support higher saturation currents due to relaxed geometric constraints on conductor cross-section, flexible winding designs for high current, and broader choices of magnetic materials and core dimensions. However, at MHz-level f_{sw}, parasitics between the inductor and the active devices increases, leading to higher interconnection losses. Moreover, the use of discrete inductors results in a larger footprint and reduced current and power density.

In contrast, embedded inductors implemented within silicon interposers, organic or glass substrates, redistribution layer (RDL) dielectrics in advanced packaging platforms, and FR-4 PCB substrates have been widely investigated [44]–[64] (see Table II). Embedded implementations enable closer coupling with switching devices, which is critical for IVRs operating at high f_{sw}, while reducing overall package dimensions and thickness. Embedding inductors eliminate off-chip interconnects and reduces parasitics, while the reduced inductor height simultaneously shortens the vertical interconnect length, thereby lowering through-via-related parasitics and PDN losses in IVPD architectures. However, embedded inductors introduce additional challenges that include fabrication complexity for three-dimensional structures, limited high-current capability, copper resistance losses, material selection, and thermal management.

Overall, the suitability of an embedded inductor with a given material and geometry for IVRs operating at high f_{sw} can be evaluated using a figure of merit (FoM) [48], defined by:

$$FoM_{ind} = L \cdot I_{Lp}^2 . f_{sw}, \tag{2}$$

where L is the inductance and I_{Lp} the peak inductor current. This metric captures the combined effects of energy-storage capability and current handling at the target switching frequency. However, in addition to the FoM, loss mechanisms play a critical role, as they directly impact efficiency and joule heating. Two inductors may exhibit similar FoM values yet demonstrate markedly different performance due to differences in dc/ac resistance and core loss as shown in Table II. For certain converter topologies, including the architecture reported in [31], the required inductance is primarily reduced through circuit-topology improvements rather than by increasing the operating frequency f_{sw}. In such cases, saturation current becomes a more critical selection criterion than the overall FoM, while core-loss characteristics also play an important role. For these converters, magnetic cores with sufficient but not excessive permeability and a gentler B–H slope, enabling higher saturation current [63], may be preferred due to the reduced inductance requirement.

The limited saturation current of fully integrated inductors can constrain the power capability of each module, necessitating multiple parallel modules to achieve kilowatt-level currents. Accordingly, [31], [46] recommends using parallel inductor networks. For example, seven parallel inductors, each rated at 5 A peak current and 14-nH inductance

Table III
Performance comparison of capacitor technologies for LV and HV embedded capacitors in integrated vertical power delivery applications

Note	Technology	Substrate	Capacitance/ Dimension (inch size)	Profile (Min. Thickness)	Maximum Voltage/ Volt. Coef. (VCC)	Capacitance Density $\mu F/mm^2$	Maximum Temperature/ Temp. Coef. (TCC)	ESR/ SRF	ESL (1GHz)	Limitations for Embedded Applications	Best-Suited Applications Based on Its Characteristics
2024, XJTU [98]	Tantalum Thin-Film	Ta Oxide-Polymer	45 μF/ 10×10 mm	50 μm	10 V/ -30 %/V	0.45	Not Reported	7.5 mΩ / 0.1 MHz	Very High	High ESL, Low Frequency Range	HV[1] Bulk ≤ 10 V
2018, GT-TI [97]	Aluminum Thin -Film	Silicon	Not Reported	110 μm	48 V/ Not Reported	2.2	125 °C/ <+4 %/°C	30 mΩ / 1 MHz	Very High	High ESL, Low Frequency Range	HV Bulk ≤ 48 V
Murata [100]-GRM031R60J	Multi-layer (MLCC)	Ceramic	0.1 μF/ 0.3×0.6 mm (0201)	100 μm	6.3 V/ -76 %/V	0.56	85 °C/ -16 %/°C	30 mΩ/ 10 MHz	190 pH	High ESL; Voltage Dependence	HV Decouple ≤ 6 V
Murata [100]-GRM011C80E	Multi-layer (MLCC)	Ceramic	0.1μF/125×250 μm (008004)	125 μm	2.5 V/ -72 %/V	3.2	105 °C/ -40 %/°C	30 mΩ/ 20 MHz	90 pH	ESL; Temperature & Voltage Dependence	LV[1] Decouple ≤ 2.5 V
Murata [100]-LLL15SD70G	Multi-layer (MLCC)	Ceramic	1 μF/ 0.5×1 mm (0402)	160 μm	4 V/ -60 %/V	2	125 °C/ -33 %/°C	5 mΩ/ 5 MHz	58 pH	ESL; Temperature & Voltage Dependence	LV Decouple ≤ 4 V
Murata [100]-GRT033D70E	Multi-layer (MLCC)	Ceramic	1 μF/ 0.3×0.6 mm (0201)	300 μm	2.5 V/ -28 %/V	5.6	125 °C/ -29 %/°C	16 mΩ/ 8 MHz	160 pH	ESL; Temperature & Voltage Dependence	LV Bulk ≤ 2.5 V
Murata [100]-GRM188C71E	Multi-layer (MLCC)	Ceramic	2.2 μF/ 0.8×1.6 mm (0603)	800 μm	25 V/ -80 %/V	1.72	125 °C/ -60 %/°C	5 mΩ/ 2.5 MHz	300 pH	Profile; ESL; Temp. /Volt. Dependence	HV Bulk ≤ 25 V
2016, Murata IPDiA [69]	Deep Trench	Silicon	0.1 μF/ 0.5×1 mm (0402)	100 μm	Not Reported	0.2	Not Reported	5 mΩ/ 90 MHz	30 pH	ESL; Restricted to Low Voltages	LV Decoupling
2022, Murata [75]	Deep Trench	Silicon	0.88 μF/1.15×1.15 mm (0404)	50 μm	Not Reported	0.67	Not Reported	5 mΩ/ 50 MHz	4 pH	Restricted to Low Voltages	LV Decoupling
2023, XMU [79]	Deep Trench-MIM	Glass	8.85 nF/ 0.36×0.36 mm	< 100 μm	7.8 V/ Not Reported	0.068	Not Reported	315 mΩ / 500 MHz	10 pH	High ESR, Ultra-Low Cap. Density	HV Decouple ≤ 7 V
2012, Philips TU/e [71],[74]	Deep Trench-MIM	Silicon	4.4 nF/ 0.1×0.1 mm	400 μm	6 V/ Not Reported	0.44	Not Reported	6 mΩ/ Not Rep.	4 pH	Low Capacitance Density	HV Decouple ≤ 6 V
2025, ELSPESED04 [78]	Deep Trench-MIM	Silicon	4.3 μF/ 1×1.2 mm	95 μm	4 V/ <+2 %/V	3.5	150 °C/ <+0.2 %/°C	< 5 mΩ/ 150 MHz	< 1 pH	Restricted to DC Voltages ≤ 4 V	LV Bulk & Decouple ≤ 4 V
2026, ELSPESED10 [77]	Deep Trench-MIM	Silicon	0.45 μF/ 1×1.2 mm	120 μm	10 V/ <+2 %/V	0.375	150 °C/ <+0.2 %/°C	< 3 mΩ/ 450 MHz	< 1 pH	Restricted to DC Voltages ≤ 10 V	HV Decoupling ≤ 10 V
2022,TIT-Dis Corp [72]	Deep Trench	Silicon Dies in Silicon	44 nF/ 3×2 mm	100 μm	Not Reported	< 0.01	Not Reported	> 65mΩ/ Not Rep.	< 1 pH	Ultra-Low Cap. Density	LV Decoupling
2026, PennStateELSPES[99]	Deep Trench-MIM	Silicon Die in Glass	2.8 μF/ 1.6×1.8 mm	210 μm	6.5 V/ <+1 %/V	1	150 °C/ <+4 %/°C	2.8 mΩ/ 140 MHz	0.45 pH	Restricted to DC Voltages ≤ 6.5 V	HV Decoupling ≤ 6 V
2026, PennStateELSPES [99]	Deep Trench-MIM	Silicon Dies in Glass	14 μF/ 1.8×8.6 mm	210 μm	6.5 V/ <+1 %/V	0.9	150 °C/ <+4 %/°C	0.5 mΩ/ 350 MHz	0.15 pH	Restricted to DC Voltages ≤ 6.5 V	HV Bulk ≤ 6 V

[1] LV and HV breakdown voltages correspond to $V \leq 5$V and $V > 5$V, respectively.

can achieve a total output current of 35-A with an equivalent inductance of 2 nH per phase, resulting in a total current of 100-A in a three-phase 5-MHz series-capacitor IVR architecture shown in Fig. 5 [31].

2) *Capacitors*

As summarized in Table I, high-voltage ratio series/flying-capacitor-based buck-derived architectures are among the most attractive candidates for high-voltage-ratio IVRs with input voltages exceeding 5 V. Unlike conventional synchronous buck IVRs and FIVRs, these architectures require not only low-voltage (LV) output capacitors, whose breakdown voltage is close to the processor supply voltage, but also high-voltage (HV) series or flying capacitors, whose breakdown voltage is close to or below the IVR input voltage.

Both LV output capacitors and HV series or flying capacitors must simultaneously satisfy high capacitance density, low parasitic ESL and ESR, and high SRF [4]. LV output capacitors directly affect processor supply noise, power integrity within the package, and overall PDN losses, thereby strongly influencing IVR performance [65]-[67]. In contrast, HV capacitors must additionally provide high breakdown voltage while maintaining a low-profile form factor [65].

Several embedded capacitor (eCAP) technologies have been explored, particularly with low ESL and ESR across a wide frequency range, while maintaining high capacitance density

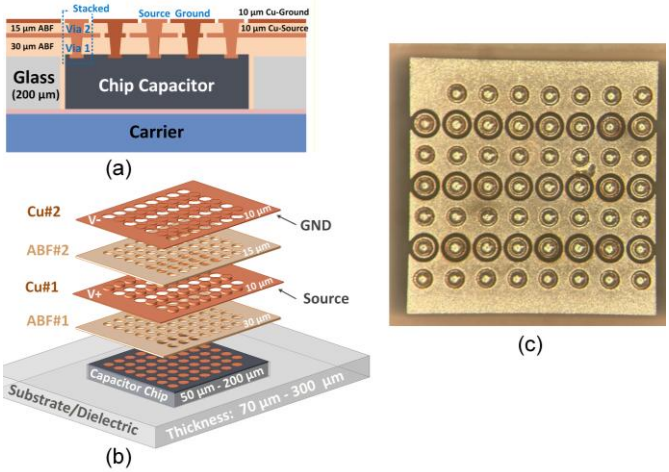


Fig. 6: (a) Embedded DTC in Glass; (b) Capacitor with multiple terminals connected to the redistribution layers (RDL) [65], [99]; and (c) fabricated structure of (b) showing top view.

and a small footprint [66]-[70]. These technologies include multilayer ceramic capacitors (MLCCs), metal–insulator–metal (MIM) capacitors, embedded film capacitors, embedded electrolytic aluminum capacitors, and silicon deep-trench capacitors (DTCs) [68], as summarized in Table III for both high (HV) and low (LV) breakdown voltage.

MLCCs are the most widely used capacitors in IVR implementations due to their high capacitance density and

broad voltage availability. MIM capacitors provide excellent high-frequency performance but are limited by relatively low capacitance density. Embedded film capacitors offer good stability but occupy larger volume, while embedded electrolytic aluminum capacitors exhibit high ESL and limited high-frequency performance, making them unsuitable for MHz-range IVRs [65], [68].

Silicon deep-trench capacitors can serve as a strong alternative to MLCCs in IVR applications. Their trench-based structure enables ultra-low-profile thicknesses, as low as approximately 50 µm, and supports multi-terminal access rather than conventional two-terminal configurations. This multi-terminal architecture, together with the low-profile form factor, facilitates seamless integration with semiconductor packaging and improves reliability [71]–[76]. In addition, silicon DTCs demonstrate excellent thermal stability, with operating temperature capability extending from 150 °C to 250 °C [76].

Despite these advantages, current silicon DTC technologies typically provide lower capacitance density than MLCCs for very high capacitance value in the tens to hundreds of microfarads range. However, for capacitance values in the nano-farad to few-microfarad range, DTCs are highly competitive and well suited for high-frequency decoupling applications [77]. Another present limitation relative to MLCCs is breakdown voltage, with commercially available DTCs commonly rated in the 5 to 10 V range, although higher-voltage variants are under active development. As a result, silicon DTCs are particularly attractive for use as output decoupling capacitors in addition to MLCCs in high-current VRM and IVR applications.

For HV capacitors required in single-stage 48 V IVRs employing series or flying capacitors, MLCCs remain the primary solution. However, in vertically stacked architectures, capacitor thickness becomes a critical constraint. In this context, the ultra-low-profile nature of DTCs makes them especially attractive, motivating continued development of high-capacitance and high-voltage DTC technologies for future high-density integrated vertical power delivery systems.

Several studies [65], [71]–[80], and [99] have reported significant progress in silicon-based DTC technology, demonstrating improvements in capacitance density, ESL, ESR, and integration with advanced packaging platforms. Reported capacitance values range from the nano-farad up to approximately 2.8 µF, with thicknesses spanning between 50 µm to 300 µm, excluding contact pads. Capacitance densities reported in the literature range from a few nano-farads per square millimeter up to 3500 nF/mm² [78], supporting applications from RF decoupling to power delivery. In [65], and [99] a 2.78 µF, 210 µm-thick silicon DTC with a 6.5-V breakdown voltage was embedded in a 200 µm-thick glass as shown in Fig. 6. The ESL of these capacitors after embedding into glass cavities connected to the RDL is around 5.6 pH along the edges of the copper planes for horizontal current paths and less than 1 pH in the vertical direction.

Overall, while MLCCs continue to dominate high-voltage series capacitor applications, silicon DTCs offer compelling advantages in profile, parasitic performance, thermal stability, and integration flexibility. Their inherently low ESL and ESR, combined with higher SRF enabled by multi-terminal trench structures, make DTCs particularly attractive. Continued advances in capacitance density and breakdown voltage are therefore expected to play a critical role in enabling future high-performance, high-density vertical power delivery architectures.

C. Multi-Physics Modeling

IVPD introduces significant challenges in modeling that include IR drop estimations, inductive noise simulation, and combined thermal-electrical modeling. IR drops up to 50 mV in multi-tier stacks and high frequency switching inductive noise exceeding 100 mV can be expected if the PDN is not designed well. Current densities up to 10 MA/cm² are also possible in vertical vias thereby intensifying electromigration and thermal hotspots. Clean and efficient IVPD requires a multi-physics modeling approach, that includes multi-scale simulations incorporating resistance-inductance-capacitance (RLC) extracted networks, for optimized via placement thereby limiting IR drop below 5% of supply voltage along with integrated decoupling capacitors for noise suppression [81], [82].

Key challenges include the following:

1) Resistive and IR drop: Arising from resistive losses in the PDN, IR drops can increase in vertical architectures due to stacked tiers. In three-dimensional heterogeneous integration (3DHI), the load is assembled at the top of the package, with power delivered through the bottom tiers. With silicon interposers, assuming a typical through silicon via (TSV) resistance of around 1 Ω and even with several TSVs in parallel, IR drops can be as high as 20-50 mV for a 1 A current and 1 V supply voltage. Moreover, dynamic IR analysis considering switching currents can result in drops up to 8% of the supply voltage for higher loads [84]. Modeling using distributed RLC networks is therefore required with TSV resistances of 20 mΩ/µm and micro-bump resistances of 5-10 mΩ. It has been shown that there is a near-linear relationship between IR drop and package height, with approximately 10 mV increase for every additional tier in the PDN. Minimizing this penalty requires careful optimization of the various PDN layers [85].

2) Inductive noise: Inductive noise, more commonly known as the Ldi/dt noise, is prominent in IVPD due to transistor switching at higher frequencies (> 1 GHz) and inductive paths. Vertical vias and micro-bumps offer mutual inductance of around 1 nH/mm resulting in voltage spikes of up to 100 mV during transients. In multi-chip packages, crosstalk from inductive coupling can exceed 8% supply noise [86]. Using full-wave electromagnetic simulations for a 48V-to-1V regulator, inductive peaks can reach up to 150 mV [83], which can be a concern unless decoupling capacitors are embedded in the package.

3) Thermal-Electrical Coupling: For AI applications, thermal gradients up to 50 °C across PDN layers is quite common. This can result in increase in resistivity by 0.4%/°C

that can increase IR drop by 20%. Capacitive coupling in dense TSV arrays can add noise, while electromigration limits current to 1 mA/TSV. Thus, multi-scale modeling (nanoscale on-chip to microscale at package level) is essential for accurate predictions [87].

An integrated method for simultaneously analyzing electrical, thermal, fluidic, and mechanical performance using 3D solvers is shown in Fig. 7. The I^2R losses incurred based on the current path through BGAs, vias, RDL interconnects, and the interposer to the active dies needs to be calculated accurately. Since significant heat is produced by the inductors and power devices under high current loads, analyzing integrated flow passages using computational fluid dynamics becomes critical for lowering the temperature of the various components. Thermal gradients in turn create mechanical stresses which get concentrated at C4 bumps, interfaces, and discontinuities. Hence, a co-design approach is necessary that can lead to high power efficiency while mitigating thermomechanical reliability risks in dense 3D packages.

Despite these challenges, simulations of multi-tier PDNs have shown that IVPD can reduce global interconnect power by nearly 40%. Studies have shown that appropriate allocation of vias for power and ground can reduce IR drops. Other studies have shown that IC switching at 3 GHz can produce 80 mV inductive noise, which can be reduced to 20 mV with embedded inductors and capacitors. An important aspect of mitigating power density and limit localized hotspots is the use of thermal vias, in that, power density per layer can drop from 21.3 W/cm² in 2D to 15.1 W/cm² in 3D IVPD. Design space exploration further indicates optimal VR placement on the interposer or substrate can reduce losses by as much as 75% [88], [89], [40].

In summary, IVPD can enhance efficiency but requires rigorous modeling to ensure IR drop and inductive noise are less than 5% and 10% of the supply voltage, respectively. Optimized IVPD can exhibit nearly 50% noise reduction if multi-physics tools are employed for sign-off. Another important design practice is to allocate vias dynamically with densities greater than 10^4/cm² to manage power and thermal integrity challenges. These modelling efforts need to be validated using full-chip models. With increased complexity of IVPD architectures, machine learning based accelerated surrogate models and multi-physics co-simulation (electrical-thermal-mechanical) will be required to achieve less than 1% error in PDN performance predictions.

D. Integrated Thermal Solution

Distributed VPD (DVPD) architectures with embedded VRs have been conceptualized for high-performance computing (HPC) systems [90]. This can offer a promising solution, but such embedded VRs can introduce substantial thermal challenges, especially in 3D stacked architectures where inner tiers have limited heat dissipation pathways. This can lead to significant heat dissipation in future technologies, which will require specialized and high-efficiency cooling solutions to manage intense heat in a confined space. The options of heat

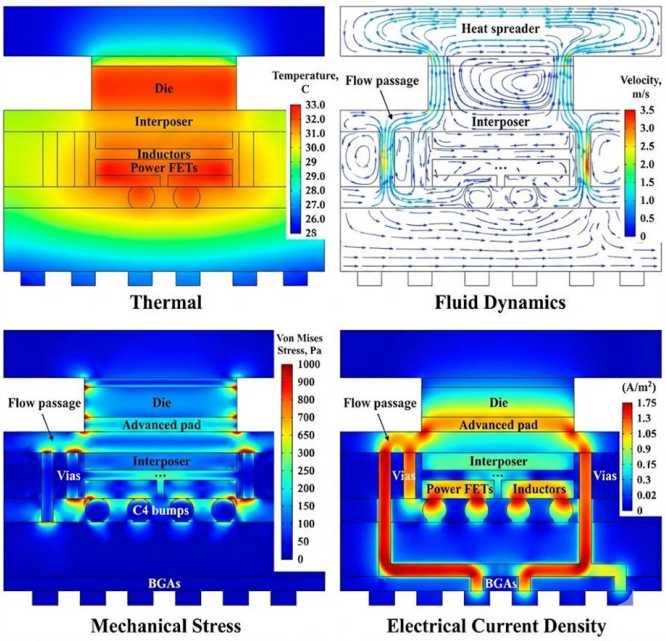


Fig. 7: Illustration of a multi-physics simulation framework that integrates electrical, thermal, and mechanical analyses using 3D solvers.

removal from these emerging IVPDs could be from the top side, bottom side, or by embedded thermal management solutions.

Top side cooling is essential for the GPU and HBM as in Fig. 3(b) and can be enabled by custom cold plates which are attached to the top of the processor, and two phase cooling (Fig. 8(a)).

An efficient cooling solution can be implemented on the bottom side for heat removal from VRs using an integrated microfluidic cold plate. One such architecture is considered in [43], where an integrated microfluidic cold plate is designed within the PWB to cool the flip-chipped GaN devices in an IVPD network (Fig. 8(c)). The cold plate consists of multiple microchannels, each 150 µm wide and high, embedded within the substrate. Simulations have shown that this approach can reduce the peak temperature of the GaN devices by > 45 °C, in comparison to cases where no cold plates are integrated.

When heat-dissipating components of VRs, such as power FETs or inductors, are embedded inside the substrate, then an embedded cooling solution may become necessary (Fig. 8(b)). Vertical PDNs have been analyzed in [90], where-in distributed in-interposer dual-phase multi-inductor hybrid converters are positioned vertically beneath the die for a 12V-to-1V single-stage conversion. The analysis shows that without embedded microchannel cooling, the maximum system temperatures reach approximately 155.5 °C, but embedded microchannels help to reduce the peak temperature below 90 °C. The analysis also establishes that by selecting appropriate channels and flowrates the power consumption for cooling can be minimized to a few tens of watts, a very small fraction compared to the heat removed by these micro-channels. In a follow-on analysis, a self-consistent electrothermal modeling framework that iteratively couples electrical losses with the thermal model can

be used to determine the required flow rates in micro-channels, and the resulting temperature distribution, leading to improved fidelity and computational efficiency [91], [93]. Analysis of air and water as coolant has shown that water cooling can achieve more than 20 K lower temperature than air cooling [92]. Embedded microchannels position the coolant very close to high power regions, leading to reduced thermal resistance between the hot component and coolant.

The advancement of data centers will demand current densities as high as 5 A/mm² and will pose significant thermal management challenges for future architectures of PDN. The embedded microchannel cooling appears to be a promising approach. An embedded microchannel with 3D manifold design can provide a more promising cooling solution as it can help in lowering the overall pressure drop in the cooling system while improving thermal performance [94], [95]. The addition of high-conductivity thermal materials replacing low-conductivity interlayer dielectrics can help with efficient heat spreading and hot spot temperature reduction in future designs. Finally, fabrication and integration of cooling solutions such as microchannels in 3DHI is still a significant challenge and requires a large size effort from both academia and industry for rapid adoption.

VII. Emerging Glass Substrate for IVPD

To illustrate the design considerations arising from architectural and passive-integration tradeoffs discussed in the previous sections, we present a representative design example of a 12-to-1 V high-frequency ZVS-BCM series-capacitor buck IVR module employing three interleaved phases (Fig. 5), on an emerging glass substrate platform [31]. Glass core combined with polymer provides good dimensional stability, low dielectric loss, high-density RDLs, enabling fine-line routing and fine-pitch vertical interconnections. Glass package also offers improved thermo-mechanical stability and reduced warpage, along with scalability to large substrate sizes and the ability to combine the interposer with the substrate. Other possibilities include silicon and organic packaging solutions which have been covered in [4] and [96].

In the representative implementation considered here, the high-side and low-side power switches and gate-driver circuitry are flip-chipped onto the bottom side of the first glass layer, the high-voltage series capacitors (C_{s1} and C_{s2} in Fig. 5) implemented using parallel discrete silicon deep-trench capacitor dies with 5 V and 10 V ratings are embedded in glass, while the output inductor and capacitor are embedded in a second glass layer above the switches. Polymer between the glass layers supports high density wiring to connect the components. The inductors can also be integrated into the RDL layers as thin films [52].

For IVRs operating at high f_{sw}, switch selection is commonly guided by device FoM, defined as:

$$FoM_{sw} = R_{DS}(on) \times Q_G, \quad (3)$$

which captures the tradeoff between drain to source resistance R_{DS} (conduction) and gate charge Q_G (gate-drive losses). For high input voltage hard-switched architectures shown in Table I in addition to FoM, the output capacitance charge Q_{oss} and P_{sw} switching losses are critical, where turn-on and capacitive losses limit f_{sw}. In ZVS-based converters, however, turn-on and Q_{oss}-related losses are largely eliminated, shifting dominance to conduction and gate-drive losses and making FoM the primary criterion for switch selection.

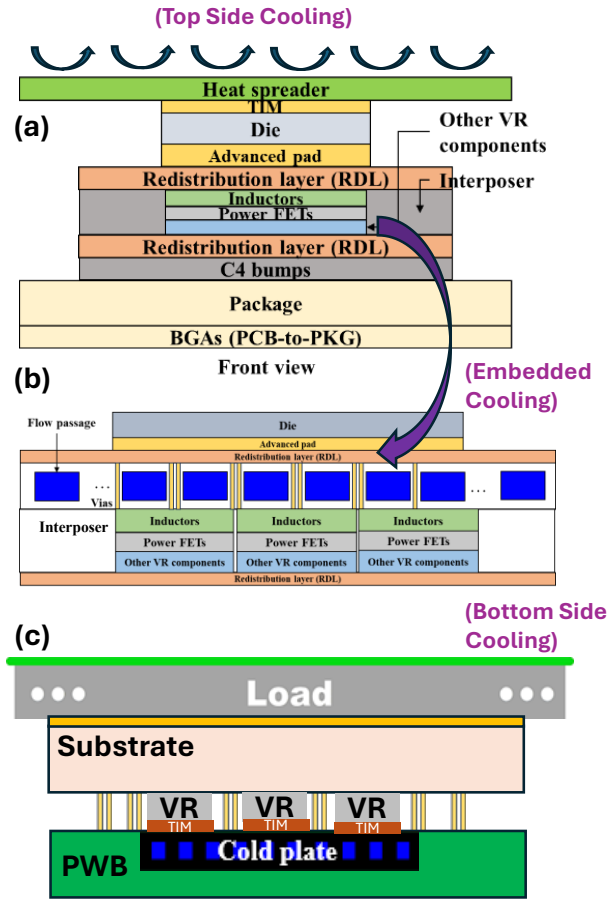


Fig. 8: (a) Example of Top Side Cooling, (b) Example of Embedded Cooling using Micro-Channels; and (c) Bottom Side Cooling using Cold Plate embedded in PWB [91].

Low-voltage GaN devices (30–60 V rating) offer favorable FoM due to their low gate charge; however, they are limited in peak current and often require device paralleling for high-current applications. While paralleling reduces the effective on-resistance of the switches, it increases the total gate charge, gate-driving losses, and package footprint, leading to a degraded system-level FoM when normalized to integration density. In contrast, commercially available low-voltage silicon MOSFETs support much higher currents, up to 789 A (e.g., Infineon IQDH29NE2LM5), with ultra-low $R_{DS}(on)$ but relatively high gate charge Q_G. Moreover, under ZVS operation, the higher output capacitance Q_{oss} and turn-on switching losses of silicon MOSFETs become less penalizing. Nevertheless, the primary limitation of silicon devices in mega-Hertz ZVS applications is their higher gate charge compared with GaN devices, which can degrade the FoM and increase gate-driving losses. However, silicon MOSFETs exhibiting sufficiently low

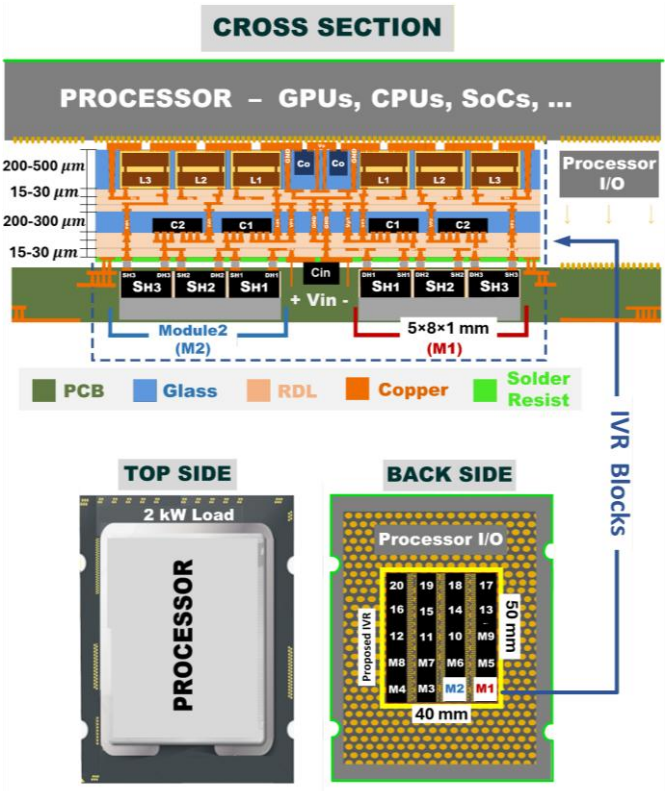


Fig. 9. IVPD implementation using glass substrates. Each IVR block contains a three-phase, 5 MHz, 12-to-1 V, ZVS-BCM buck converter, with S_{H1}–S_{H3} and S_{L1}–S_{L3} in Fig. 5 serving as the high-side and low-side switches, respectively, placed on the same substrate layer. The gate drivers are located in close proximity to the switches. The HV-series capacitors (C_{S1} and C_{S2}) are embedded in the first glass layer, while the composite/thin-film inductors with embedded windings and a profile of <0.5 mm [50], [60] are integrated into the second layer. The RDL thickness is determined by the required routing layers. Twenty identical modules on a 40 × 50 mm substrate, providing a total output current of 2000 A to the top-side load.

Q_G can still provide competitive performance, especially in high-current applications where their superior current capability offsets the gate-drive penalty. Therefore, the EPC2366 GaN device (ultra-low FOM, ~88 A peak limit) and the Infineon BSC009NE2LS silicon series (slightly higher FoM but much higher current capability) are used as representative switches for the loss analysis in this section.

To achieve a total output power of 2000 W at 1 V, multiple parallel modules (or distributed VRs) are required, since available active and passive components cannot individually support such high current levels, and delivering this power using a single module with parallel components is not practical. For this architecture three interleaved phases represent the maximum practical phase count for 12-to-1 V conversion to achieve low output current ripple, whereas up to six interleaved phases can be employed for 48-to-1 V conversion, as discussed in [31], [32]. Considering the three-phase configuration and the peak current limitations of GaN switches, the maximum power per module is limited to approximately 100 W. Consequently, achieving a total output power of 2000 W requires operating at least 20 interleaved modules, each employing three-phase series-capacitor stages in parallel with details in Table IV and Fig. 9.

Table IV.
Converter Parameters for a 2-kW, 5-MHz, 12-to-1-V Design Based on the Maximum Current Limitations of Commercial GaNs.

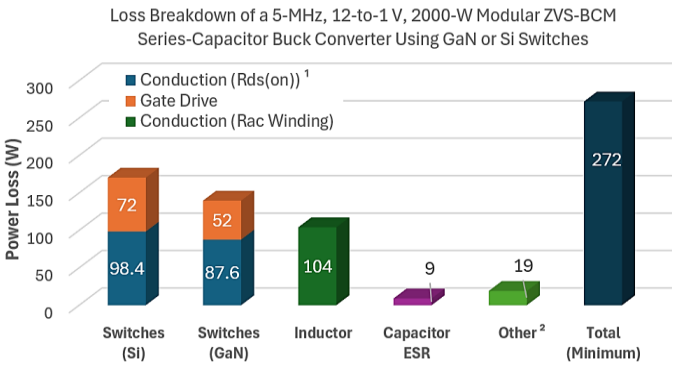
12-1 V 5 MHz	Switch Type		No. Module	Po: Total / Module	Inductance / I_{rms} (Per Phase)	Full-Load Efficiency
Design 1 (High-Q air-core)	GaN	EPC 2366	20	2 kW / 100 W @ 1V	1.85 nH / 41 A	88%
	Si	BSC 009				86.8%

For the series capacitor architectures in [31], in BCM operation, the inductor current exhibits large ripples, causing the rms current to approach the peak value. As a result, inductor copper loss is dominated by ac resistance (skin effect for air-core) rather than dc resistance, making the inductor quality factor (Q) at the operating f_{sw} an important design parameter.

For this example, an air-core inductor network with an equivalent inductance of approximately 1.85 nH is selected for loss breakdown. The network is modeled using parallel elements to support 41 A rms per phase in a 100 W three-phase module exhibiting a quality factor of approximately 58 at 5 MHz, corresponding to a total ac resistance of $R_{ac} \approx 1\ \text{m}\Omega$. As a replacement, parallel configurations of high-saturation-current magnetic-core inductors to further reduce ac copper losses and enhance overall efficiency are required.

Fig. 10 presents the overall loss breakdown for a representative 5 MHz, 12-to-1 V, 2000 W ZVS-BCM series-capacitor buck converter using best-available technologies. Switch conduction and gate-drive losses account for a moderate fraction of the total loss when GaN switches are selected, while capacitor ESR loss remains relatively small due to the use of high-quality, embedded, ultra-low-ESL multi-terminal deep-trench capacitors. In contrast, the inductor represents the largest loss contributor.

The results indicate that the ZVS architecture, by effectively eliminating turn-on switching losses, enables high-f_{sw} operation at high input voltage through negligible Q_{oss} and turn-on switching losses ($P_{sw}(on)$). However, for this architecture, the realization of suitable embedded passive components, particularly inductors, remains the dominant barrier to achieving conversion efficiencies above 90%. In this example, ultra-low ac resistance (R_{ac}) is achieved by paralleling multiple air-core inductors, resulting in 88% full-load efficiency. High-FoM silicon switches can also achieve relatively high efficiency in ZVS-based architectures due to negligible turn-on switching and capacitive losses; however, GaN devices provide superior performance because of their lower gate-charge losses. Specifically, the use of a high-FoM silicon MOSFET in the ZVS-BCM architecture (Table IV) reduces efficiency by approximately 1.4% compared with the state-of-the-art GaN device.



[1] $R \times I_{rms}^2$ losses for all switches: Rds(on) @ (80 ℃, VGS = 5 V, ID=50 A)
[2] Other losses include interconnection and T_{dead} losses of the low-side switches.

Fig. 10: Loss breakdown of a 5-MHz, 12-to-1 V, 2000-W ZVS three-phase series-capacitor buck converter [31], comparing the minimum achievable losses using optimized GaN and Si switches and high-Q air-core inductor.

VIII. Conclusion

In conclusion, emerging substrate technologies provide unique opportunities to improve power delivery efficiencies through integrated vertical power delivery (IVPD). However, a few technological advancements are still required to fully realize high-efficiency, kW-scale, IVPD solutions. These include:

a. Low-voltage GaN devices (≤ 40 V) with reduced $R_{DS}(on)$ and higher current capability are required. While existing GaN devices offer favorable FOM, their limited current capability necessitates device paralleling, which increases gate-drive losses and package footprint, leading to a degraded system-level FOM when normalized to integration density. For ZVS architectures, higher $P_{sw}(on)$ switching and Q_{oss} parameters are tolerable, providing design freedom for future high-current, low-voltage GaN devices targeted at data-center power delivery applications.

b. Embedded inductors with magnetic core, low loss tangent and high saturation current are needed. Moderate core permeability is sufficient for ZVS-BCM architecture if it enables high saturation current in a low-profile structure, while achieving reduced ac copper loss. Alternatively, very high-Q air-core inductors with extremely low inductance and high current capability can also be considered.

c. High-breakdown-voltage low-profile embedded capacitors are required. For high-conversion ratio VRs employing series and flying capacitors, current deep-trench capacitor technologies are limited to breakdown voltages of approximately 10 V. Future development must focus on extending DTC voltage ratings toward 50 V while preserving the ultra-low-profile advantage over commercial MLCCs.

d. With large substrate composites, coefficient of thermal expansion (CTE) that minimizes mechanical stresses at the chip-package and package-PWB are required along with warpage control to ensure reliability especially for large size substrates of the order of 150mm – 200mm on a side. Multi-physics modeling plays an important role in ensuring that the mechanical stresses are addressed while electrical and thermal needs are met.

e. Power delivery with GPU, HBM, embedded chips such as CPUs and integrated inductors require robust cooling solutions using a combination of single phase and two-phase cooling.

f. Architectural-level co-optimization and advanced design-automation are required. Scalable IVPD deployment demands system-level electrical–thermal–mechanical co-design and reduced-order multi-physics modeling frameworks capable of managing the computational complexity of multi-VR, large-substrate architectures.

References

[1] E. Sperling, “Crisis Ahead: Power Consumption in AI Data Centers,” SemiEngineering, Jul. 21, 2025. [Online]. Available: https://semiengineering.com/crisis-ahead-power-consumption-in-ai-data-centers. [Accessed: Feb. 12, 2026].

[2] M. Swaminathan, J. Kim, I. Novak and J. P. Libous, "Power distribution networks for system-on-package: status and challenges," in *IEEE Transactions on Advanced Packaging*, vol. 27, no. 2, pp. 286-300, May 2004, doi: 10.1109/TADVP.2004.831897.

[3] M. Blake, M. Hsu, I. Goldwasser, H. Petty, and J. Huntington, “NVIDIA 800 VDC Architecture Will Power the Next Generation of AI Factories,” NVIDIA Technical Blog, May 20, 2025. [Online]. Available: https://developer.nvidia.com/blog/nvidia-800-v-hvdc-architecture-will-power-the-next-generation-of-ai-factories/. [Accessed: Feb. 12, 2026].

[4] K. Radhakrishnan, M. Swaminathan and B. K. Bhattacharyya, "Power Delivery for High-Performance Microprocessors—Challenges, Solutions, and Future Trends," in *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 11, no. 4, pp. 655-671, April 2021, doi: 10.1109/TCPMT.2021.3065690.

[5] S. Carlo, W. Yueh and S. Mukhopadhyay, "On the potential of 3D integration of inductive DC-DC converter for high-performance power delivery," *2013 50th ACM/EDAC/IEEE Design Automation Conference (DAC)*, Austin, TX, USA, 2013, pp. 1-8, doi: 10.1145/2463209.2488955.

[6] M. Halamicek, T. McRae and A. Prodić, "Cross-Coupled Series-Capacitor Quadruple Step-Down Buck Converter," *2020 IEEE Applied Power Electronics Conference and Exposition (APEC)*, New Orleans, LA, USA, 2020, pp. 1-6, doi: 10.1109/APEC39645.2020.9124412.

[7] Y. Zhu, J. Zou and R. C. N. Pilawa-Podgurski, "A 1500-A/48-V-to-1-V Switching Bus Converter for Next-Generation Ultra-High-Power Processors," in *IEEE Transactions on Power Electronics*, vol. 39, no. 9, pp. 11340-11355, Sept. 2024, doi: 10.1109/TPEL.2024.3403670.

[8] M. Ursino, S. Saggini, S. Jiang and C. Nan, "High density 48V-to-PoL VRM with hybrid pre-regulator and fixed-ratio buck," *2020 IEEE Applied Power Electronics Conference and Exposition (APEC)*, New Orleans, LA, USA, 2020, pp. 498-505, doi: 10.1109/APEC39645.2020.9124561.

[9] Bel-Power, Main & Satellite Power Stamp 48V-to-PoL Isolated DC-DC Converters, https://www.belfuse.com/media/datasheets/products/power-supplies/ds-bps-48-v-to-pol-power-stamp.pdf

[10] Z. Ye, R. A. Abramson, Y. -L. Syu and R. C. N. Pilawa-Podgurski, "MLB-PoL: A High Performance Hybrid Converter for Direct 48 V to Point-of-Load Applications," *2020 IEEE 21st Workshop on Control and Modeling for Power Electronics (COMPEL)*, Aalborg, Denmark, 2020, pp. 1-8, doi: 10.1109/COMPEL49091.2020.9265870.

[11] Y. Zhu, T. Ge, Z. Ye and R. C. N. Pilawa-Podgurski, "A Dickson-Squared Hybrid Switched-Capacitor Converter for Direct 48 V to Point-of-Load Conversion," *2022 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Houston, TX, USA, 2022, pp. 1272-1278, doi: 10.1109/APEC43599.2022.9773567.

[12] R. Das, G. -S. Seo, D. Maksimovic and H. -P. Le, "An 80-W 94.6%-Efficient Multi-Phase Multi-Inductor Hybrid Converter," *2019 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Anaheim, CA, USA, 2019, pp. 25-29, doi: 10.1109/APEC.2019.8721952.

[13] *Analog Devices*, 54VIN Dual 25A, Single 50A µModule Regulator with Digital Power System Management, LTM4664, Jan. 2021, rev. B.

[14] Y. Chen *et al*., "Virtual Intermediate Bus CPU Voltage Regulator," in *IEEE Transactions on Power Electronics*, vol. 37, no. 6, pp. 6883-6898, June 2022, doi: 10.1109/TPEL.2021.3130213.

[15] J. Zhu and D. Maksimovic, "48 V-to-1 V Transformerless Stacked Active Bridge Converters with Merged Regulation Stage," *2020 IEEE 21st Workshop on Control and Modeling for Power Electronics (COMPEL)*, Aalborg, Denmark, 2020, pp. 1-6, doi: 10.1109/COMPEL49091.2020.9265671.

[16] M. H. Ahmed, C. Fei, F. C. Lee and Q. Li, "Single-Stage High-Efficiency 48/1 V Sigma Converter With Integrated Magnetics," in *IEEE Transactions on Industrial Electronics*, vol. 67, no. 1, pp. 192-202, Jan. 2020, doi: 10.1109/TIE.2019.2896082.

[17] N. M. Ellis and R. C. N. Pilawa-Podgurski, "A Symmetric Dual-Inductor Hybrid Dickson Converter for Direct 48V-to-PoL Conversion," *2022 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Houston, TX, USA, 2022, pp. 1267-1271, doi: 10.1109/APEC43599.2022.9773452.

[18] P. R. Prakash *et al*., "A 2400 W/in3 1.8 V Bus Converter Enabling Vertical Power Delivery for Next-Generation Processors," *2024 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Long Beach, CA, USA, 2024, pp. 910-917, doi: 10.1109/APEC48139.2024.10509453.

[19] J. Baek *et al*., "Vertical Stacked LEGO-PoL CPU Voltage Regulator," in *IEEE Transactions on Power Electronics*, vol. 37, no. 6, pp. 6305-6322, June 2022, doi: 10.1109/TPEL.2021.3135386.

[20] O. Kirshenboim and M. M. Peretz, "High-Efficiency Nonisolated Converter With Very High Step-Down Conversion Ratio," in *IEEE Transactions on Power Electronics*, vol. 32, no. 5, pp. 3683-3690, May 2017, doi: 10.1109/TPEL.2016.2589321..

[21] N. M. Ellis, Y. Zhu and R. C. N. Pilawa-Podgurski, "Gallium Nitride-based 48V-to-1V Point-of-Load (PoL) Converter for Aerospace Telecommunications and Computing Applications," *2024 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Long Beach, CA, USA, 2024, pp. 1384-1388, doi: 10.1109/APEC48139.2024.10509202.

[22] *Vicor*, PRM™ Regulator High Efficiency Converter, PRM48BH480x250A00, Nov. 2020, rev. 1.7.

[23] *Vicor*, VTM™ Current Multiplier Sine Amplitude Converter™ (SAC™), VTM48MP010x107AA1, Jul. 2017, rev. 1.5.

[24] Y. Elasser *et al*., "Mini-LEGO: A 1.5-MHz 240-A 48-V-to-1-V CPU VRM with 8.4-mm Height for Vertical Power Delivery," *2023 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Orlando, FL, USA, 2023, pp. 1959-1966, doi: 10.1109/APEC43580.2023.10131163.

[25] M. Gong, X. Zhang and A. Raychowdhury, "A 90.4% Peak Efficiency 48V/1V Three-Level Hybrid Dickson Converter with Gradient Descent Run-Time Optimizer and GaN/Si Hybrid Conversion," *2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, Honolulu, HI, USA, 2022, pp. 176-177, doi: 10.1109/VLSITechnologyandCir46769.2022.9830284.

[26] P. S. Shenoy," Introduction to the Series Capacitor Buck Converter" *Texas Instruments* DC Solutions Application Report, SLVA750A–April 2016–Revised May 2016.

[27] P. S. Shenoy *et al*., "A 5 MHz, 12 V, 10 A, monolithically integrated two-phase series capacitor buck converter," *2016 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Long Beach, CA, USA, 2016, pp. 66-72, doi: 10.1109/APEC.2016.7467853.

[28] M. Gong *et al*., "A GaN-Based Reconfigurable Series-Parallel Hybrid Converter Supporting 48/24/12V Input and 0.8-1.2V Output with 83.7/87.8/90.7% Peak Efficiency," *2023 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Orlando, FL, USA, 2023, pp. 912-918, doi: 10.1109/APEC43580.2023.10131253.

[29] S. Ren *et al*., "A Two-Stage 12–1-V Converter Featuring Regulated Resonant SC Regulators and Collaborative Control Scheme," in *IEEE Journal of Solid-State Circuits*, vol. 60, no. 12, pp. 4369-4382, Dec. 2025, doi: 10.1109/JSSC.2025.3609744.

[30] S. Amin *et al*., "A 5.4V-Vin, 9.3A/mm2 10MHz Buck IVR Chiplet in 55nm BCD Featuring Self-Timed Bootstrap and Same-Cycle ZVS Control," *2024 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, Honolulu, HI, USA, 2024, pp. 1-2, doi: 10.1109/VLSITechnologyandCir46783.2024.10631523.

[31] R. R. Khorasani, X. Li, J. W. Kim, P. Murali, R. Sharma and M. Swaminathan, "Package Power Delivery Architecture for High Performance Computing Systems With a 1 kW IVR Operated in CCM-DCM Boundary Mode Condition," *2024 IEEE 74th Electronic Components and Technology Conference (ECTC)*, Denver, CO, USA, 2024, pp. 285-292, doi: 10.1109/ECTC51529.2024.00054.

[32] R. R. Khorasani, K. P. Ghitelman and M. Swaminathan, "Design and Implementation of a GaN-Based Soft-Switched Series-Capacitor Buck Converter Operating at the CCM-DCM Boundary for High-Performance Computing Systems," *2025 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Atlanta, GA, USA, 2025, pp. 2101-2108, doi: 10.1109/APEC48143.2025.10977125.

[33] H. Ren *et al*., "Power Delivery for Scaled-Out Chiplet-Based Wafer-Scale Systems with 8µm Cu-Cu Bond Pitch on Active SI-Interconnect Fabric Substrate," *2025 Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, Kyoto, Japan, 2025, pp. 1-3, doi: 10.23919/VLSITechnologyandCir65189.2025.11074799.

[34] C. Schaef *et al*., "8.5 A Fully Integrated Voltage Regulator in 14nm CMOS with Package-Embedded Air-Core Inductor Featuring Self-Trimmed, Digitally Controlled Variable On-Time Discontinuous Conduction Mode Operation," *2019 IEEE International Solid-State Circuits Conference - (ISSCC)*, San Francisco, CA, USA, 2019, pp. 154-156, doi: 10.1109/ISSCC.2019.8662294.

[35] W. Gomes *et al*., "Ponte Vecchio: A Multi-Tile 3D Stacked Processor for Exascale Computing," *2022 IEEE International Solid-State Circuits Conference (ISSCC)*, San Francisco, CA, USA, 2022, pp. 42-44, doi: 10.1109/ISSCC42614.2022.9731673.

[36] M. Gong, X. Zhang and A. Raychowdhury, "Non-isolated 48V-to-1V Heterogeneous Integrated Voltage Converters for High Performance Computing in Data Centers," *2020 IEEE 63rd International Midwest Symposium on Circuits and Systems (MWSCAS)*, Springfield, MA, USA, 2020, pp. 411-414, doi: 10.1109/MWSCAS48704.2020.9184442.

[37] B. Soleymani, O. Bagheri, E. Adib and S. Eren, "A Coupled Inductor High Step-Down Voltage Regulator Module With Zero Current Soft-Switching," in *IEEE Transactions on Power Electronics*, vol. 40, no. 11, pp. 16664-16675, Nov. 2025, doi: 10.1109/TPEL.2025.3586796.

[38] M. Chen, S. Jiang, J. A. Cobos and B. Lehman, "Design Considerations for 48-V VRM: Architecture, Magnetics, and Performance Tradeoffs," *2023 Fourth International Symposium on 3D Power Electronics Integration and Manufacturing (3D-PEIM)*, Miami, FL, USA, 2023, pp. 1-9, doi: 10.1109/3D-PEIM55914.2023.10052608.

[39] K. Bharath *et al*., "Integrated Voltage Regulator Efficiency Improvement using Coaxial Magnetic Composite Core Inductors," *2021 IEEE 71st Electronic Components and Technology Conference (ECTC)*, San Diego, CA, USA, 2021, pp. 1286-1292, doi: 10.1109/ECTC32696.2021.00208.

[40] S. Krishnakumar *et al*., "Vertical Power Delivery for High Performance Computing Systems with Buck-Derived Regulators," *2024 IEEE 74th Electronic Components and Technology Conference (ECTC)*, Denver, CO, USA, 2024, pp. 2136-2142, doi: 10.1109/ECTC51529.2024.00364.

[41] S. Krishnakumar and I. Partin-Vaisband, "Vertical Power Delivery for Emerging Packaging and Integration Platforms - Power Conversion and Distribution," *2023 IEEE 36th International System-on-Chip Conference (SOCC)*, Santa Clara, CA, USA, 2023, pp. 1-6, doi: 10.1109/SOCC58585.2023.10256973.

[42] S. Krishnakumar *et al*., "Design Considerations for DC-DC Voltage Regulators in Distributed Vertical Power Delivery Systems," *2024 IEEE International Symposium on Circuits and Systems (ISCAS)*, Singapore, Singapore, 2024, pp. 1-5, doi: 10.1109/ISCAS58744.2024.10558456.

[43] M. Choi, S. Krishnakumar, R. Rahimzadeh Khorasani, M. Swaminathan, I. Partin-Vaisband and S. Kumar, "Substrate-Embedded Microfluidic Cooling of Distributed Vertical Power Delivery Architectures for High-Performance Computing Processors," in *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 15, no. 9, pp. 1912-1920, Sept. 2025, doi: 10.1109/TCPMT.2025.3544105.

[44] Y. Dou, Z. Ouyang, P. Thummala and M. A. E. Andersen, "PCB embedded inductor for high-frequency ZVS SEPIC converter," *2018 IEEE Applied Power Electronics Conference and Exposition (APEC)*, San Antonio, TX, USA, 2018, pp. 98-104, doi: 10.1109/APEC.2018.8340994.

[45] C. . -L. Chen *et al*., "Ultra-low-resistance 3D InFO inductors for integrated voltage regulator applications," *2016 IEEE International Electron Devices Meeting (IEDM)*, San Francisco, CA, USA, 2016, pp. 35.2.1-35.2.4, doi: 10.1109/IEDM.2016.7838546.

[46] W. J. Lambert, M. J. Hill, K. Radhakrishnan, L. Wojewoda and A. E. Augustine, "Package Inductors for Intel Fully Integrated Voltage Regulators," in *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 6, no. 1, pp. 3-11, Jan. 2016, doi: 10.1109/TCPMT.2015.2505665.

[47] D. Yu, "A new integration technology platform: Integrated fan-out wafer-level-packaging for mobile applications," *2015 Symposium on VLSI Technology (VLSI Technology)*, Kyoto, Japan, 2015, pp. T46-T47, doi: 10.1109/VLSIT.2015.7223697.

[48] P. Murali, *Materials and Processes for High Conversion Ratio High Efficiency Package Embedded Inductors for Integrated Voltage Regulators*, Ph.D. dissertation, School of Materials Science and Engineering, Georgia Institute of Technology, Atlanta, GA, USA, Dec. 2023, ProQuest no. 32315870.

[49] C. Alvarez *et al*., "Design and Demonstration of Single and Coupled Embedded Toroidal Inductors for 48V to 1V Integrated Voltage Regulators," *2020 IEEE 70th Electronic Components and Technology Conference (ECTC)*, Orlando, FL, USA, 2020, pp. 405-413, doi: 10.1109/ECTC32862.2020.00072.

[50] C. Alvarez Barros *et al*., "Embedded Inductors Using Composite Magnetic Materials for 12–1-V Integrated Voltage Regulators," in *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 11, no. 12, pp. 2183-2192, Dec. 2021, doi: 10.1109/TCPMT.2021.3116946.

[51] M. Sankarasubramanian *et al*., "Magnetic Inductor Arrays for Intel® Fully Integrated Voltage Regulator (FIVR) on 10th generation Intel® Core™ SoCs," *2020 IEEE 70th Electronic Components and Technology Conference (ECTC)*, Orlando, FL, USA, 2020, pp. 399-404, doi: 10.1109/ECTC32862.2020.00071.

[52] P. Murali *et al*., "Fabrication and Characterization of Package Embedded Inductors for Integrated Voltage Regulators," *2022 IEEE 72nd Electronic Components and Technology Conference (ECTC)*, San Diego, CA, USA, 2022, pp. 301-305, doi: 10.1109/ECTC51906.2022.00056..

[53] T. Fukuoka *et al*., "An 86% Efficiency, 20MHz, 3D-Integrated Buck Converter with Magnetic Core Inductor Embedded in Interposer Fabricated by Epoxy/Magnetic-Filler Composite Build-Up Sheet," *2019 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Anaheim, CA, USA, 2019, pp. 1561-1566, doi: 10.1109/APEC.2019.8722209.

[54] R. Murphy, Z. Pavlovic, P. McCloskey, C. O Mathuna, S. O'Driscoll and G. Weidinger, "PCB Embedded Toroidal Inductor for 2MHz Point-of-Load Converter," *CIPS 2020; 11th International Conference on Integrated Power Electronics Systems*, Berlin, Germany, 2020, pp. 1-5.

[55] H. T. Le *et al*., "High-Q Three-Dimensional Microfabricated Magnetic-Core Toroidal Inductors for Power Supplies in Package," in *IEEE Transactions on Power Electronics*, vol. 34, no. 1, pp. 74-85, Jan. 2019, doi: 10.1109/TPEL.2018.2847439..

[56] N. Sturcken, R. Davies, C. Cheng, W. E. Bailey and K. L. Shepard, "Design of coupled power inductors with crossed anisotropy magnetic core for integrated power conversion," *2012 Twenty-Seventh Annual IEEE Applied Power Electronics Conference and Exposition (APEC)*, Orlando, FL, USA, 2012, pp. 417-423, doi: 10.1109/APEC.2012.6165853..

[57] S. L. Selvaraj *et al*., "On-Chip Thin Film Inductor for High Frequency DC-DC Power Conversion Applications," *2020 IEEE Applied Power Electronics Conference and Exposition (APEC)*, New Orleans, LA, USA, 2020, pp. 176-180, doi: 10.1109/APEC39645.2020.9124544.

[58] N. Sturcken *et al*., "Magnetic thin-film inductors for monolithic integration with CMOS," *2015 IEEE International Electron Devices Meeting (IEDM)*, Washington, DC, USA, 2015, pp. 11.4.1-11.4.4, doi: 10.1109/IEDM.2015.7409676.

[59] A. Roth *et al*., "Heterogeneous Power Delivery for 7nm High-Performance Chiplet-Based Processors using Integrated Passive Device and In-Package Voltage Regulator," *2020 IEEE Symposium on VLSI Technology*, Honolulu, HI, USA, 2020, pp. 1-2, doi: 10.1109/VLSITechnology18217.2020.9265105.

[60] V. Avula, P. Murali and M. Swaminathan, "Design and Demonstration of Dual-Core Spiral Package-Embedded Inductors for Integrated Voltage Regulators," in *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 15, no. 6, pp. 1275-1283, June 2025, doi: 10.1109/TCPMT.2025.3557745.

[61] G. Zhou *et al*., "High Inductance Density Glass Embedded Inductors for 3-D Integration," in *IEEE Microwave and Wireless Technology Letters*, vol. 33, no. 6, pp. 679-682, June 2023, doi: 10.1109/LMWT.2023.3248146.

[62] P. Pan, C. Chen, J. Gu, M. Liu and X. Li, "High-Performance 3-D Silicon-Embedded Coupled Solenoid Inductors With Inserted Magnetic Core," in *IEEE Transactions on Electron Devices*, vol. 71, no. 5, pp. 3169-3174, May 2024, doi: 10.1109/TED.2024.3381920.

[63] P. Murali, P. Bhaskar, C. C. Buchanan, D. A. Gilbert, M. Swaminathan and M. D. Losego, "Effect of Metal Filler Modification on Toroidal Inductors for Package Embedded Integrated Voltage Regulators," in *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 15, no. 10, pp. 2170-2177, Oct. 2025, doi: 10.1109/TCPMT.2025.3604372.

[64] R. Rasheedi and I. Partin-Vaisband, "High Aspect Ratio Spiral Inductor with Progressive Turn Widths for Embedded Power Converters," *2025 IEEE 75th Electronic Components and Technology Conference (ECTC)*, Dallas, TX, USA, 2025, pp. 2271-2277, doi: 10.1109/ECTC51687.2025.00386.

[65] R. R. Khorasani, X. Li, M. Al-Juwhari, W. Murti, J. Cha and M. Swaminathan, "Embedded Silicon Chip Capacitors in Glass Package for Vertical Power Delivery," *2025 IEEE 75th Electronic Components and Technology Conference (ECTC)*, Dallas, TX, USA, 2025, pp. 416-423, doi: 10.1109/ECTC51687.2025.00076..

[66] R. Tummala and M. Swaminathan, *System on Package: Miniaturization of the Entire System*, vol. 1. New York, NY, USA: McGraw-Hill, 2007.

[67] P. Muthana *et al*., "Analysis of Embedded Package Capacitors for High Performance Components," *2006 IEEE Electrical Performane of Electronic Packaging*, Scottsdale, AZ, USA, 2006, pp. 55-58, doi: 10.1109/EPEP.2006.321189..

[68] Spurney, R. G., *Advanced Materials and Processes for High-Density Capacitors for Next-Generation Integrated Voltage Regulators*, Ph.D. dissertation, Georgia Institute of Technology, 2019.

[69] C. Bunel and F. Murray, "Ultra Thin, Low ESL and High Frequency Performance of High Density Silicon Capacitors," in Proc. 2016 *Int. Symp. Microelectron*., Pasadena, CA, USA, Oct. 2016, pp. 60–63.

[70] J. Phillips et al., "Determining the Requirements, Die vs. Package vs. Board: Multi-level Power Distribution Network Design," presented at the *DesignCon 2025 Conf.*, Santa Clara, CA, USA, Jan. 2025.

[71] J. H. Klootwijk *et al*., "Ultrahigh Capacitance Density for Multiple ALD-Grown MIM Capacitor Stacks in 3-D Silicon," in *IEEE Electron Device Letters*, vol. 29, no. 7, pp. 740-742, July 2008, doi: 10.1109/LED.2008.923205. .

[72] K. Kobinata *et al*., "Low-ESL (<1 pH @ 8.5 GHz) Multi-Terminal Si Capacitor Embedded in 3D Functional Interposer for Power Delivery Network," *2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, Honolulu, HI, USA, 2022, pp. 385-386, doi: 10.1109/VLSITechnologyandCir46769.2022.9830411.

[73] S. Banzhaf, S. Schwaiger, T. Erlbacher, A. J. Bauer, L. Frey and L. Frey, "Post-trench processing of silicon deep trench capacitors for power electronic applications," *2016 28th International Symposium on Power Semiconductor Devices and ICs (ISPSD)*, Prague, Czech Republic, 2016, pp. 399-402, doi: 10.1109/ISPSD.2016.7520862.

[74] M. K. Matters-Kammerer, K. B. Jinesh, T. G. S. M. Rijks, F. Roozeboom and J. H. Klootwijk, "Characterization and Modeling of Atomic Layer Deposited High-Density Trench Capacitors in Silicon," in *IEEE Transactions on Semiconductor Manufacturing*, vol. 25, no. 2, pp. 247-254, May 2012, doi: 10.1109/TSM.2012.2183903.

[75] M. M. Jatlaoui *et al*., "Multi-terminal Ultra-thin 3D Nanoporous Silicon Capacitor Technology for High-Speed Circuits Decoupling," *2022 IEEE 72nd Electronic Components and Technology Conference (ECTC)*, San Diego, CA, USA, 2022, pp. 908-913, doi: 10.1109/ECTC51906.2022.00148.

[76] Bunel, C., and L. Lengignon, "Silicon Capacitors with Extremely High Stability and Reliability Ideal for High Temperature Applications," *IMAPSource Proceedings*, vol. 2012, HITEC, pp. 273–277, 2012.

[77] ELSPES, "Silicon Capacitor vs. MLCC," [Online]. Available: https://elspes.com/technology/silicon-capacitor-vs-mlcc. [Accessed: Jan. 31, 2026].

[78] ELSPES, " ELSPES Inc. Achieves Breakthrough 3,500 nF/mm² Cap. Density in Silicon DTC," [Online]. Available: https://www.semiconductorpackagingnews.com/uploads/2/2025_ELSPES_News_3500_nF.pdf [Accessed: January 31, 2026].

[79] Z. Hu *et al*., "Development of Low Cost Glass-Based Deep Trench Capacitor for 3D Packaging," in *IEEE Electron Device Letters*, vol. 44, no. 9, pp. 1535-1538, Sept. 2023, doi: 10.1109/LED.2023.3295596.

[80] S. Gandhi, M. R. Pulugurtha, H. Sharma, P. Chakraborti and R. R. Tummala, "High-k Thin-Film Capacitors With Conducting Oxide Electrodes on Glass Substrates for Power-Supply Applications," in *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 6, no. 10, pp. 1561-1566, Oct. 2016, doi: 10.1109/TCPMT.2016.2606434.

[81] P. Jain, P. Zhou, C. H. Kim, and S. S. Sapatnekar, "Thermal and power delivery challenges in 3D ICs," in Three Dimensional Integrated Circuit Design: EDA, Design and Microarchitectures, Y. Xie, J. Cong, and S. S. Sapatnekar, Eds. Boston, MA, USA: Springer, 2009, pp. 33–61, doi: 10.1007/978-1-4419-0784-4_3.

[82] K. Chang, S. Das, S. Sinha, B. Cline, G. Yeric and S. K. Lim, "System-Level Power Delivery Network Analysis and Optimization for Monolithic 3-D ICs," in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 27, no. 4, pp. 888-898, April 2019, doi: 10.1109/TVLSI.2019.2897589.

[83] Z. Li, Y. Ma, Q. Zhou, Y. Cai, and Y. Xie, "Thermal-aware P/G TSV planning for IR drop reduction in 3D ICs," *Integration*, vol. 46, pp. 1–9, 2013.

[84] H. He, Z. Xu, X. Gu and J. -Q. Lu, "Power delivery modeling for 3D systems with non-uniform TSV distribution," *2013 IEEE 63rd Electronic Components and Technology Conference*, Las Vegas, NV, USA, 2013, pp. 1115-1121, doi: 10.1109/ECTC.2013.6575713.

[85] I. A. Papistas and V. F. Pavlidis, "Crosstalk noise effects of on-chip inductive links on power delivery networks," *2016 IEEE International Symposium on Circuits and Systems (ISCAS)*, Montreal, QC, Canada, 2016, pp. 1938-1941, doi: 10.1109/ISCAS.2016.7538953.

[86] Z. Chen, J. Zhang, S. Wang, and C.-P. Wong, "Challenges and prospects for advanced packaging," *Fundamental Research*, vol. 4, no. 6, pp. 1455–1458, 2024, doi: 10.1016/j.fmre.2023.04.014.

[87] Hao Yu, Joanna Ho, and Lei He, "Allocating Power Ground Vias in 3D ICs for Simultaneous Power and Thermal Integrity," *ACM Transactions on Design Automation of Electronic Systems* (TODAES), Volume 14, Issue 3, Pages 1 – 31, 2009.

[88] W. Ahmad, "*Core Switching Noise for On-Chip 3D Power Distribution Networks*," Ph.D. dissertation, School of Information and Communication Technology, KTH Royal Institute of Technology, Stockholm, Sweden, 2012.

[89] S. J. Souri, "3D ICs Interconnect Performance Modeling and Analysis," Ph.D. dissertation, Stanford University, Stanford, CA, USA, 2003.

[90] M. Choi *et al*., "Thermal Analysis of High Current Vertical Power Delivery Network with Embedded Microchannel Cooling," *2024 23rd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm)*, Aurora, CO, USA, 2024, pp. 1-8, doi: 10.1109/ITherm55375.2024.10709419.

[91] M. Choi *et al*., "Self-Consistent Electrothermal Modeling of Distributed Vertical Power Delivery Architecture With Substrate-Embedded Microfluidic Cooling," in *IEEE Transactions on Components, Packaging and Manufacturing Technology*, doi: 10.1109/TCPMT.2025.3636391.

[92] F. Xie and T. Wei, "Exploring Efficient Thermal Management Solutions for Backside Power Delivery Network (BSPDN) Systems Using Multiscale Modeling," in *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 15, no. 6, pp. 1237-1247, June 2025, doi: 10.1109/TCPMT.2025.3551225.

[93] F. Xie, R. Chen and T. Wei, "Thermal Mitigation Strategy for Backside Power Delivery Network," *2024 IEEE 74th Electronic Components and Technology Conference (ECTC)*, Denver, CO, USA, 2024, pp. 1485-1492, doi: 10.1109/ECTC51529.2024.00241.

[94] Z. Wu, X. Zhang, S. Tian, A. R. Kidambi, J. A. Weibel and L. Pan, "Modeling of Backside Power Delivery and Thermal Management in Semiconductor Die Packages," *2024 23rd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm)*, Aurora, CO, USA, 2024, pp. 1-7, doi: 10.1109/ITherm55375.2024.10709612.

[95] A. Piazza *et al*., "Considerations and Challenges for Large Area Embedded Micro-channels with 3D Manifold in High Heat Flux Power Electronics Applications," *2020 19th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm)*, Orlando, FL, USA, 2020, pp. 77-82, doi: 10.1109/ITherm45881.2020.9190179.

[96] S. Ravichandran & M. Swaminathan, "Heterogeneous Integration for AI Applications: Status & Future Needs, *Chip Scale Review,* 2022.

[97] R. Spurney et al., "Demonstration of Hermetic Sealing on Ultra-Thin, Wafer-Integrated Aluminum-Polymer Capacitors for High-Voltage and High-Temperature Applications," 2018 IEEE 68th Electronic Components and Technology Conference (ECTC), San Diego, CA, USA, 2018, pp. 1414-1420.

[98] J. Zhao, S. Liu, R. Zhao, J. An and Y. Xu, "Design and Implementation of High-Performance Tantalum Thin-Film Capacitors for Embedded Technologies," in *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 15, no. 4, pp. 868-873, April 2025, doi: 10.1109/TCPMT.2025.3543402

[99] R. R. Khorasani et al., "Fabrication and Experimental Characterization of Embedded Multi-Terminal Capacitors With Ultra-Low Parasitics for Integrated Vertical Power Delivery," 2026 IEEE 76th Electronic Components and Technology Conference (ECTC), Orlando, FL, USA, 2026, pp. 723-730, doi: 10.1109/ECTC51846.2026.00123.

[100] Murata Manufacturing Co., Ltd., *Design Support Software "SimSurfing"*. [Online]. Available: https://www.murata.com/en-us/tool/simsurfing. Accessed: Jun. 15, 2026.



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